Predicting Open-Market Residential Property Values in Metro Cebu Using Machine Learning and Geospatial Features

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By

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Approval Sheet

The capstone entitled **Predicting Open-Market Residential Property Values in Metro Cebu Using Machine Learning and Geospatial Features**, prepared and submitted by **Chris Dominic Estreba** in partial fulfillment of the requirements of the degree of **Bachelor of Science in Data Science**, is approved.

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Abstract

This study developed a residential property valuation framework for Metro Cebu using machine learning and GIS-derived spatial features. The work addressed the gap between administrative benchmarks, lender-driven appraisals, and market-facing price signals by assembling a dataset of open-market residential listings collected from three online property portals. The study covered six local government units: Cebu City, Mandaue City, Lapu-Lapu City, Talisay City, Minglanilla, and Consolacion.

The methodology combined structural, administrative, and geospatial variables. Property records were geocoded, matched with barangay-level BIR zonal values, and enriched with accessibility and neighborhood-context features, including proximity to major economic nodes, amenity access, and a neighborhood price lag. Because property types are priced on very different per-square-meter logics, separate models were fitted for condominiums, houses, and vacant lots. Within each stratum, hedonic regression, Random Forest, and XGBoost were compared under leak-free grouped cross-validation. The tree-based models outperformed the hedonic baseline and performed comparably with each other, and Random Forest was deployed in all three strata for its robustness on small samples.

The deployed models reached a typical error of about 19%, 23%, and 38% (median absolute percentage error) for condominiums, houses, and vacant lots, with location features carrying most of the predictive signal; accuracy was lower and more variable for vacant lots. Outputs were interpreted through SHAP-based explainability and delivered through a prototype web-based decision-support application centered on a predicted residential price surface. The contribution is a locally grounded valuation prototype that combines property-level machine learning, spatial feature engineering, and transparent explanation. Given the remaining error, especially for vacant lots, it is positioned as a prototype for triangulation rather than a stand-alone valuation tool, though it is already more responsive to urban change than static administrative benchmarks alone.

Acknowledgment

This page is reserved for the acknowledgment.

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List of Abbreviations and Acronyms

ABT	Analytics Base Table
APA	American Psychological Association
BIR	Bureau of Internal Revenue
BSP	Bangko Sentral ng Pilipinas
CBD	Central Business District
CBRT	Cebu Bus Rapid Transit
CRISP-DM	Cross-Industry Standard Process for Data Mining
GIS	Geographic Information System
IVS	International Valuation Standards
LGU	Local Government Unit
MCRAI	Metro Cebu Residential Accessibility Index
OLS	Ordinary Least Squares
OSM	OpenStreetMap
POI	Point of Interest
ROPA	Real and Other Properties Acquired
RPPI	Residential Real Estate Price Index
SHAP	SHapley Additive exPlanations

Introduction

Background of the Study

What is Real Estate?

Real estate sits at the intersection of shelter, commerce, and investment. Under Article 415 of the Philippine Civil Code, immovable property includes land, buildings, and permanent attachments. Unlike many other assets, real estate is both something people use and something people must price, finance, sell, tax, and sometimes defend as collateral.

In practice, a property price is rarely decided by one actor alone. A seller needs a reasonable listing range. A buyer needs to know whether an asking price is defensible. Brokers and agents bring local market knowledge and comparable evidence into negotiations. Banks assess collateral risk. Appraisers estimate value under professional standards. Local government units set administrative benchmarks such as zonal values for taxation. These actors do not all ask the same question, but they all depend on evidence that can make a price range understandable.

Real Estate in Cebu

Cebu has become one of the Philippines' most dynamic regional economies, recording 7.3% growth in 2024 (Philippine Statistics Authority – Central Visayas, 2025). Much of this growth has come from the IT-BPM sector, the expansion of call center activity, and the post-pandemic recovery of tourism, hospitality, and retail. These shifts have fed directly into demand for residential and mixed-use property.

Across Metro Cebu, particularly Cebu City, Mandaue, Lapu-Lapu, Talisay, Minglanilla, and Consolacion, that demand has made the property market notably active. Residential property prices in the area rose by 11.5% in 2025, the highest rate outside the National Capital Region (Bangko Sentral ng Pilipinas, 2025).

That pressure is also being reinforced by infrastructure projects that are changing accessibility across the urban area. The Cebu Bus Rapid Transit (CBRT), the Metro Cebu Expressway, and the continued development of the South Road Properties are likely to affect land values by altering connectivity, travel times, and the attractiveness of nearby locations. For a valuation study, these spatial shifts matter because they change not only where demand is strongest, but also how quickly older reference points can become less informative.

The Problem: How is Price Decided in Cebu?

Despite this growth, a seller or buyer in Metro Cebu still does not have a simple public source that says what a specific residential property is likely to command in the current market. Instead, pricing is usually assembled from several partial reference points. Each one is useful, but each one answers a different question.

First, broker and agent opinions remain important because they reflect local knowledge of subdivisions, barangays, comparable properties, buyer inquiries, and negotiation conditions. The limitation is not the absence of expertise. The limitation is that this evidence is not always standardized, transparent, or reproducible across practitioners, especially in markets where transaction data are thin or difficult to access. Studies of developing-country valuation problems similarly point to limited and unreliable market information as a major source of valuation difficulty, rather than simply individual professional failure (Ajibola, 2010; Cheloti & Mooya, 2021).

Second, bank appraisals provide formal valuation evidence, but their purpose is tied to lending and collateral risk. A value produced for loan security is useful, but it does not necessarily answer the seller's question of what listing range the current market may tolerate. Third, BIR zonal values and local assessment schedules provide official administrative references, but they are designed primarily for taxation and public assessment. BIR zonal values are not live transaction prices (Bureau of Internal Revenue, n.d.), and Philippine evidence shows that local assessment schedules are not always updated regularly (Otsuka et al., 2023). Fourth, online listings are current and property-level, but they are asking prices rather than verified sale prices, so they contain seller strategy, negotiation buffers, and platform-level noise.

The result is not that existing references are useless. Rather, the problem is that they are difficult to compare because they serve different purposes, follow different assumptions, and update at different speeds. For buyers and sellers, this makes price ranges harder to interpret. For brokers and appraisers, it limits the availability of a common evidence base for explaining judgments. For banks, it complicates collateral assessment. For local governments, it weakens the connection between administrative benchmarks and market-facing price evidence. Metro Cebu therefore has active valuation activity, but still lacks a transparent, spatially detailed reference layer for interpreting property-level residential price evidence.

Why Metro Cebu, and Why Now?

Three converging factors make it timely to build a data-driven decision-support layer for Metro Cebu:

1. **Prices are moving faster than slow references can follow**: When market-facing prices move quickly, administrative benchmarks and older comparable evidence become harder to use without adjustment.

2. **Infrastructure is redrawing the value map**: Projects like the CBRT and the Expressway are restructuring barangay accessibility and desirability. Spatial changes matter because two properties with similar structural features can occupy very different market positions depending on access to employment centers, roads, amenities, and surrounding price levels.

3. **Market-facing and geospatial data can now be organized more systematically**: Online platforms such as Lamudi, open geospatial data, and geocoding tools enable the construction of a property-level dataset without requiring private deed-of-sale records. This does not make the data equivalent to verified transactions, but it does allow market-facing asking-price evidence to be mapped, modeled, and compared more transparently.

Ideal Scenario

The core problem is not that Cebu lacks valuation activity, professional judgment, or local market knowledge. The problem is that these judgments are often made against fragmented references. An ideal decision-support layer would organize observable property attributes, location, nearby amenities, benchmark values, and surrounding market-facing prices in one transparent framework. Such a layer would not replace brokers, banks, appraisers, or government benchmarks. Instead, it would help users see where a property sits relative to observable market evidence and why a model places weight on certain value drivers.

This study therefore does not claim to produce a final appraised market value. It develops a transparent, geospatial decision-support model for interpreting market-facing residential asking-price evidence in Metro Cebu. Delivered through a web application, the model is intended as an additional reference layer that supports professional review and transparent comparison, consistent with valuation-standard concerns around data, inputs, and model explainability (International Valuation Standards Council, 2025).

Statement of the Problem

Decision Problem: How can real estate stakeholders in Metro Cebu compare residential property price evidence more transparently, when broker judgment, bank valuation, government benchmarks, and online listings each serve a different purpose?

Research Problem: These references are difficult to compare at the property level, and no Cebu-specific study identified in the reviewed literature combines property-level asking-price data with geospatial features, such as network distance, MCRAI scoring, and spatial autocorrelation, while also providing a spatial decision-support layer for valuation practice.

Research Questions

1. What value drivers significantly influence residential property prices in Metro Cebu?

2. Among **Hedonic Regression**, **Random Forest**, and **XGBoost**, which model is most suitable for deployment when balancing out-of-sample accuracy, robustness on small per-stratum samples, and interpretability?

3. Do **geospatial features**—proximity to economic nodes, MCRAI accessibility measures, and spatial autocorrelation—improve valuation accuracy over a structural-only model, and how does that contribution differ across property types?

4. How large is the valuation gap between model-implied market-facing residential price estimates and traditional BIR Zonal Values across Metro Cebu?

Significance of the Study

Addressing this fragmented evidence problem benefits multiple stakeholders:

1. **Real Estate Brokers and Appraisers**: Provides an additional evidence layer for comparing local market knowledge against structured property, location, and benchmark data. SHAP-based explainability and a web interface can help make the model's reasoning easier to inspect during professional review.

2. **Property Sellers, Buyers, and Investors**: Provides a transparent reference for interpreting whether an asking price appears high, low, or typical relative to observable market-facing evidence. It does not determine the final selling price, but it can make the starting point for discussion more evidence-based.

3. **Banks and Lending Institutions**: Offers market-facing spatial evidence on how location and surrounding market conditions relate to residential price, which complements rather than substitutes for collateral-oriented valuation.

4. **Local Government Units (LGUs)**: Maps the divergence between market-facing asking-price evidence and administrative benchmarks, which can help identify where benchmark schedules may require closer review.

This study is significant within the Philippine context because it develops a Cebu-specific, property-level valuation workflow that integrates GIS-based feature engineering with explainable machine learning. While prior Philippine machine learning studies focus on Metro Manila or broader indicator-based approaches, no work identified in the reviewed literature applies geocoded network-distance features, MCRAI scoring, and spatial autocorrelation to property-level residential asking-price data for Metro Cebu. By delivering results through a web application, the study extends predictive analytics into applied decision support for a fragmented local market.

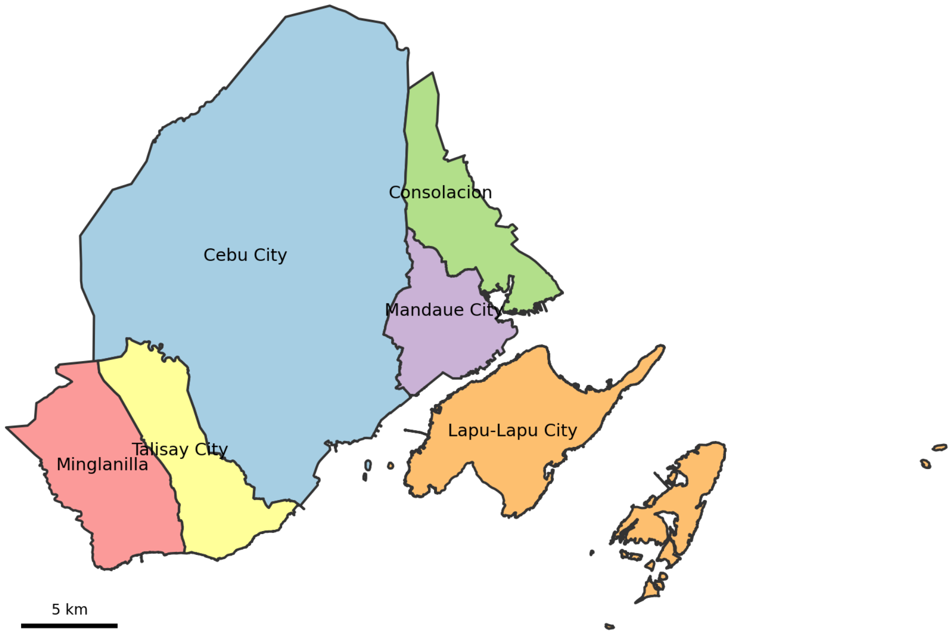
Scope and Limitations

Scope

This study focuses on the residential real estate market within Metro Cebu. While the National Economic and Development Authority (NEDA) defines Metro Cebu as encompassing 13 LGUs, this study operationally scopes “Metro Cebu” to its central urban core: **Cebu City, Mandaue, Lapu-Lapu, Talisay, Minglanilla, and Consolacion**. This constraint supports high data density and robust geospatial analysis, as these six LGUs represent the most active transaction zones and the highest concentration of infrastructure development. Map 1 shows the spatial extent of this scope.

The study drew on open-market residential listings from several online property portals, BIR zonal values, geospatial APIs, and road network data. Each source is described in detail in Chapter 3, Section 3.2.

The study evaluated three predictive models: **Hedonic Regression (OLS)**, **Random Forest**, and **XGBoost** (see Section 1.6 for rationale). To respect the very different price-per-square-meter logics of different property types, the models were fitted separately for condominiums, houses, and vacant lots. Geospatial features were engineered through geocoding, road-network distance, MCRAI scoring, and spatial lag computation. The fitted models were trained on open-market residential listings so that the deployed output stayed aligned with market-facing asking-price evidence.



Map 1

Metro Cebu study area: the six local government units considered.

The study produced an applied deliverable: a prototype web application combining an open-market residential price surface with property-level prediction and explanation. Because the model still carries substantial error for some property types, the application is presented as a prototype for triangulation rather than a tool for confident stand-alone valuation.

Limitations

1. **Open-Market Listing Bias**: The deployed model was fitted on open-market listings rather than verified deed-of-sale transactions. This kept the prediction target aligned with market-facing pricing, but it also meant that seller strategy and asking-price noise remained in the data.

2. **Unobserved Heterogeneity**: The model cannot capture qualitative features omitted from listings, such as interior finish quality or views, typically assessed during physical inspections.

3. **Geospatial Approximation**: Addresses are geocoded via Google Maps API, which may introduce minor spatial errors compared to exact lot-parcel shapefiles. Additionally, OSM coverage in peripheral barangays may be less robust than in the urban core.

4. **Temporal Scope**: The analysis provides a cross-sectional snapshot as of late 2025; it does not capture long-term cyclical real estate trends.

5. **Geographic Coverage**: The six selected LGUs represent the urban conurbation but exclude the broader municipalities of Cebu Province.

Model Selection Rationale

This study employs three complementary models that span the interpretability–accuracy spectrum. These model families are common in comparable real estate prediction studies and provide a practical basis for comparing an interpretable hedonic baseline with non-linear tree-based alternatives (Pai & Wang, 2020; Yacim & Boshoff, 2018).

- **Hedonic Regression (OLS)** serves as the interpretable baseline rooted in hedonic pricing theory (Rosen, 1974). OLS produces explicit coefficients for each value driver, making it useful for transparent comparison even when it is not the strongest predictive model. The baseline is extended with a spatial lag term to account for autocorrelation among neighboring properties.

- **Random Forest** (Breiman, 2001) addresses the linearity assumption of OLS by capturing non-linear interactions between value drivers—particularly relevant for Metro Cebu housing data, where structural and locational effects are unlikely to remain linear. Its bagging mechanism also provides robustness against outliers.

- **XGBoost** (T. Chen & Guestrin, 2016) was included as a strong candidate for predictive accuracy: gradient boosting consistently performs well on structured tabular data (Grinsztajn et al., 2022) and integrates natively with SHAP (Lundberg & Lee, 2017) to provide property-level explainability, which supports transparent professional review under valuation-standard concerns. Chapter 7 compares these models using accuracy, robustness, and deployment suitability rather than treating a single error metric as the only basis for selection.

Why Not Other Models?

Several alternatives were evaluated but ultimately set aside:

Table 2

Models Considered but Not Retained in the Study Design

Model	Rationale for Exclusion
Deep Neural Networks	Require >10,000 labeled samples; the dataset size is insufficient, and global interpretability is sacrificed.
Support Vector Regression	Computationally expensive to tune; poor native interpretability; generally weaker than boosting on tabular data.
LASSO / Ridge	Constrained by linearity assumptions; effectively subsumed by the OLS baseline for this study.

Review of Related Literature

Purpose

This chapter reviews the existing body of knowledge on data-driven property valuation, drawing from both Philippine and international literature. It establishes the theoretical foundations underpinning this study, surveys empirical evidence on machine learning approaches to real estate pricing, and identifies a critical research gap: the absence of a Cebu-specific, property-level, ML-augmented valuation model.

The review moves from valuation concepts and the Philippine setting to the practical constraints that shape valuation work in emerging markets. It then turns to modeling approaches, the use of geospatial features in property valuation, the scope of macroeconomic factors, and the compliance issues that matter if these models are to be used in practice. The chapter closes by identifying the specific gap that this study addresses in the Metro Cebu context.

Core Concepts and Philippine Context

Valuation Foundations

Market value is the most probable price under fair, open-market conditions on the valuation date. Market price is the amount actually paid in a specific transaction; the two can differ in practice (Philippine Valuation Standards Council, 2018). Hedonic Pricing Theory, introduced by Rosen (1974), provides the foundational framework for this study. It posits that the price of a differentiated good—such as a house—can be decomposed into a function of its constituent attributes: $P = f(\text{Structural}, \text{Locational}, \text{Environmental})$. This theory underpins the hedonic regression model used as our interpretive baseline.

Philippine Indices and Administrative Benchmarks

The Bureau of Internal Revenue (BIR) issues zonal values primarily for tax purposes, such as capital gains tax (CGT) and documentary stamp tax (DST). These values are administratively set and are not live market prices (Bureau of Internal Revenue, n.d.). Critically, the Tax Policy Study of 2023 (TPS 2023) found that approximately 60% of Local Government Units (LGUs) have not updated their Schedules of Market Values within the prescribed cycle. Both these local schedules and the BIR zonal values therefore lag actual market prices in many areas, including parts of Cebu.

For market-level context, the Bangko Sentral ng Pilipinas (BSP) publishes the Residential Real Estate Price Index (RREPI), which reported a 7.5% nationwide increase in housing prices in Q2 2025. Metro Cebu posted 11.5%—one of the highest growth rates outside the National Capital Region—reflecting sustained demand driven by IT-BPM, tourism, and infrastructure development (Bangko Sentral ng Pilipinas, 2025).

Cebu-Specific Empirical Work

Agosto (2020) conducted the only Cebu-specific empirical study on land value determinants, surveying 51 real estate practitioners. The study identified transport accessibility as the primary driver of land values, followed by neighborhood quality and environmental conditions. However, the study was survey-based and did not utilize property-level geocoded data or machine learning methods. That leaves a gap between what Cebu practitioners identify as important and what has actually been tested using property-level data and predictive models.

The Core Problem: Data Scarcity in Emerging Markets

International literature consistently identifies data scarcity—not valuer misconduct—as the primary obstacle to accurate property valuation in developing countries.

- **Kenya**: Cheloti and Mooya (2021) conducted a census survey of 427 registered valuers in Nairobi. "Limited information" was ranked as the primary valuation problem (Mean Rank 2.91), while valuer misconduct ranked last (2.32). They concluded that limited and unreliable information, rather than incompetence, causes valuation problems.

- **Nigeria (Lagos)**: Ajibola (2010) surveyed 300 valuers and found that 92.7% cited insufficient market evidence as the primary challenge. Valuation inaccuracy ranged from +24.8% to +51.5%, far exceeding the global norm of $\pm 10\%$.

Taken together, these studies suggest that valuation error in emerging markets is often rooted less in professional failure than in thin, inconsistent, or inaccessible market data. That framing matters for this study because it shifts attention from individual appraisal practice to the quality, structure, and availability of the information on which valuation depends.

Modeling Approaches: Traditional vs. Machine Learning

Hedonic Regression

Classical hedonic regression models decompose property price into a function of structural and locational attributes, typically in a log-linear form (Malpezzi, 2003; Rosen, 1974). Spatial econometrics extends this framework by incorporating neighborhood effects and spatial autocorrelation (Anselin, 1988). For the Philippines, Dann et al. (2020) applied hedonic and spatial models to Metro Manila data (2000–2010) and found that structural variables, environmental quality, and spatial spillovers all significantly explain price variation.

Philippine Machine Learning Studies

More recent Philippine work incorporates machine learning:

- **Viray (2023)** compared Multiple Linear Regression with Random Forest for predicting property prices in Central Pangasinan, combining BIR zonal values, BSP RPPI, and construction cost indices. The tree-based model achieved a lower prediction error.

- **Ramolete et al. (2023)** demonstrated that incorporating government indicators improves ML valuation accuracy, achieving a MAPE of 10–21% with larger, cleaner datasets.

- **Perdio et al. (2023)** tested linear models against gradient boosting on Manila listings, finding gradient boosting superior after feature selection.

International Evidence: The Tanzania Benchmark

The most directly relevant international study is Nyanda et al. (2024), who tested eight ML algorithms on approximately 954 residential properties in Dar es Salaam—a sample size close to that of the individual property strata in this study (condominiums 1,300, houses 1,223, and vacant lots 849).

Table 3

Dar es Salaam Benchmark Results Reported by Nyanda et al. (2024)

Model	MAPE
Neural Network	108.6% (failed due to overfitting on small data)
Random Forest	52.7%
XGBoost	**48.0%**

Given a sample size close to that of this study, the Tanzania study is useful less as a fixed benchmark than as a caution about model choice under limited data conditions. In that setting, tree-based models performed more reliably than deep learning architectures, which were more prone to overfitting. For a dataset of roughly 1,000 observations, the evidence points toward Random Forest and XGBoost as more defensible starting points than neural networks.

Southeast Asian Applications

Work from Southeast Asia points in a similar direction. In Indonesia, Tanamal et al. (2023) used a Random Forest model to sort Surabaya houses into price categories from a set of agent-sourced attributes, including land and building area, number of rooms, road width, building age, and area classification, and reported that the tree-based approach handled this mixed attribute data well. In Malaysia, Ja'afar and Mohamad (2021) showed that Random Forest best captured the value contribution of heritage characteristics for pre-war shophouses in Penang, outperforming more conventional regression approaches. These studies come from different property contexts, but together they suggest that tree-based models are well suited to the varied attribute data that residential valuation depends on in fast-changing urban markets.

Comparative Reviews

Review studies suggest that this is not an isolated pattern. Grinsztajn et al. (2022) found that tree-based ensembles such as Random Forest and XGBoost remain highly competitive with, and often outperform, deep learning on structured tabular data. Moreno-Foronda et al. (2025) and Sharma et al. (2024) likewise report strong performance from XGBoost in house price prediction tasks. SHAP-based explainability (Lundberg & Lee, 2017), meanwhile, can narrow the interpretability gap between less transparent ML models and more conventional hedonic approaches.

Geospatial Feature Engineering in Property Valuation

For residential property, location is not a secondary attribute. It shapes accessibility, neighborhood quality, exposure to surrounding land uses, and connection to the wider urban system. Tobler's First Law of Geography, that near things tend to be more related than distant ones (Tobler, 1970), provides a clear basis for treating spatial relationships as part of the valuation problem rather than as background context.

Geocoding and Location Precision

Geocoding matters in this study because the dataset begins with imperfect addresses rather than parcel-level coordinates. Converting those addresses into latitude and longitude makes the later proximity, amenity, and neighborhood calculations possible. Services such as the Google Maps Geocoding API are widely used for this purpose because they can handle incomplete or inconsistently formatted addresses while still producing coordinates precise enough for urban spatial analysis (Google LLC, 2025). In the present study, Google Maps served as the primary geocoder for property records, while the OpenStreetMap road graph supported later routing and network-distance calculations.

For open-source alternatives, **OpenStreetMap (OSM)** via the Nominatim geocoder provides community-maintained geospatial data at no cost. While OSM coverage varies by region, urban areas in the Philippines—particularly Metro Cebu—have benefited from coverage by the local OpenStreetMap community and humanitarian mapping initiatives (Humanitarian OpenStreetMap Team, 2024). This study employs Google Maps API as the primary geocoding engine for address-to-coordinate conversion, while OpenStreetMap supports the road-network layer used in accessibility computation.

Proximity Analysis and Accessibility

Distance-based features—proximity to commercial centers, transportation hubs, schools, and employment nodes—are robust value drivers in hedonic pricing literature (Malpezzi, 2003; Rosen, 1974). While straight-line distance remains a common baseline in urban studies (Sinnott, 1984), current valuation workflows increasingly rely on road-network distance because it better reflects actual accessibility. This thesis follows that later approach, using network distance as the standard measure with Haversine distance only as a fallback when routing is unavailable.

Agosto (2020) confirmed that transport accessibility is the primary driver of land values in Cebu, followed by neighborhood quality. For Metro Cebu, proximity to polycentric economic nodes—including Cebu Business Park, Mandaue, Mactan, South Road Properties, Tabunok, Consolacion, Naga City, and the airport area—provides measurable value signals that are more defensible than a monocentric CBD assumption.

Accessibility Indexing and OpenStreetMap Support

Beyond direct point-to-point distances, the density and diversity of nearby amenities provides a richer description of neighborhood quality. The Python library osmnx enables programmatic access to OSM data for both network analysis and amenity retrieval, and has been used in property valuation research to generate proximity features from road networks and point-of-interest counts (Boeing, 2017, 2019).

In the Philippine setting, the use of OpenStreetMap cannot simply be assumed. Alvarez et al. (2021), through Project OHANA, provide more locally relevant support for its use by showing that OSM data can sustain nationwide amenity accessibility analysis in the country. Their work does not eliminate data quality concerns, but it does suggest that OSM is sufficiently robust to support spatial analysis of the kind required in this study. In the final thesis workflow, that support role is paired with external point-of-interest inventories rather than treated as the sole amenity source.

Spatial Autocorrelation and Neighbor Price Effects

A critical consideration in property valuation is **spatial autocorrelation**: properties located near each other tend to have similar prices, violating the independence assumption of standard regression (Anselin, 1988). This phenomenon is well-documented: housing prices exhibit positive spatial dependence because neighboring properties share the same schools, transportation access, environmental quality, and market conditions.

Two approaches address spatial effects in modeling:

1. **Spatial Lag Model (SLM)**: Includes a spatially weighted average of neighboring property prices as an independent variable, directly capturing the effect of nearby market conditions on a property's value.

2. **Moran's I Statistic**: A diagnostic measure of global spatial autocorrelation (Moran, 1950). A statistically significant positive Moran's I in the residuals of a non-spatial model indicates that spatial effects are present and should be incorporated.

This study incorporates a **spatial lag variable**—the mean price of properties within a defined radius—as a feature in the ML models. This operationalizes Tobler's law within the predictive framework without imposing the parametric constraints of formal spatial econometric models.

Implications for Metro Cebu

This matters in Metro Cebu because value shifts are unlikely to be evenly distributed across the urban area. Infrastructure projects such as the CBRT and Metro Cebu Expressway, uneven amenity access, and strong variation across neighborhoods all create spatial differences that a purely structural model would miss. In that sense, geospatial feature engineering is not an optional enhancement but a way of making the valuation model more responsive to how the city is actually changing.

Integrating Value Drivers: Indexing vs. Raw Features

A critical challenge in geospatial ML models is how to operationalize distance metrics. Agosto (2020) identified transport accessibility and neighborhood quality as the primary value drivers in Cebu based on practitioner surveys. To integrate these findings into a machine learning pipeline, raw geospatial distances (e.g., Euclidean distance to the nearest hospital, school, or mall) are often insufficient on their own, as they can suffer from multicollinearity—a scenario where multiple distance variables are highly correlated with one another, complicating the model's feature importance weights.

Recent work suggests that raw distance variables are often too fragmented to stand on their own. Rey-Blanco et al. (2024), for example, show that accessibility indices built from point-of-interest data can improve housing price prediction in both hedonic regression and Random Forest models. Grouping individual distances and counts into broader measures such as transit accessibility or commercial density can preserve the locational signal while reducing overlap across variables. For this study, that approach provides a practical way to translate Agosto's (2020) qualitative value drivers into features that can be used consistently in the model.

Macroeconomic Factors and Study Scope

Broad macroeconomic conditions, such as exchange rates, inflation, and interest rates, influence the direction of a housing market over time. This study does not model those forces. Its design is cross-sectional: it explains differences in price per square meter across locations at a single point in time, using property attributes and spatial features rather than market-wide cycles. Macroeconomic determinants therefore fall outside the scope of the present model. They remain a sensible direction for later work, and Chapter 10 records them as a recommendation for further research.

Compliance and Explainability: IVS 2025

The International Valuation Standards (2025), effective January 31, 2025, introduced two critical new chapters relevant to data-driven valuation:

- **IVS 104 (Data and Inputs)**: Mandates that data used in valuations must be Accurate, Complete, Timely, and Transparent. Sources must be traceable from their origin ("provenance requirement").
- **IVS 105 (Valuation Models)**: Explicitly states that *"No model without the valuer applying professional judgement can produce an IVS-compliant valuation."* Automated Valuation Models (AVMs) must be tested, transparent, and paired with professional review.

For this study, the IVS revisions matter in two practical ways.

1. **SHAP Values for Transparency**: This study employs SHAP (SHapley Additive exPlanations) to provide both global feature importance ("Which value drivers affect Metro Cebu prices most?") and local explanations ("This Mactan condominium is higher-valued because of stronger accessibility and locational advantages, but lower floor area pulls the estimate down."). This satisfies IVS 104's transparency requirement.
2. **Human-in-the-Loop Validation**: Licensed real estate brokers from the CPRE network are engaged to review model outputs, satisfying IVS 105's professional judgement mandate. This makes the model a *decision-support tool*, not a replacement for the appraiser.

Synthesis and Research Gap

Taken together, the literature points to four recurring themes. First, valuation work in emerging markets is persistently constrained by data scarcity and uneven market information. Second, for small to mid-sized tabular datasets, tree-based machine learning models tend to perform more reliably than deep learning approaches. Third, spatial features such as proximity, accessibility measures, and neighborhood effects add information that conventional structural variables alone do not capture. Fourth, any model intended for valuation practice still has to remain transparent enough to support professional review under IVS 2025.

What remains missing is a Cebu-specific study that brings those strands together using property-level data. Prior Philippine ML studies focus on Metro Manila (Dann et al., 2020; Perdio et al., 2023), Central Pangasinan (Viray, 2023), or broader indicator-based approaches rather than property-level modeling (Ramolete et al., 2023). The only Cebu-specific study identified here, Agosto (2020), is valuable in showing what practitioners regard as important, but it is survey-based and does not test those factors using observed property data.

This thesis addresses that gap by developing a Metro Cebu valuation model that combines open-market listings from multiple online portals, tree-based machine learning, geospatial feature engineering, SHAP-based explainability, and professional review. The aim is not to replace appraisal judgement, but to produce a more transparent and spatially grounded decision-support tool for the local market.

Bridge to Methodology

Chapter 3 builds on this review by showing how those ideas are translated into data sources, engineered features, model comparisons, and evaluation procedures for Metro Cebu. In other words, the next chapter moves from what the literature suggests to how those insights are operationalized in this study.

Methodology

Research Design

This study used a quantitative, non-experimental design to model residential property values in Metro Cebu. The analysis was predictive in that it estimated prices from observed property characteristics, and prescriptive in that the results were later organized for spatial decision support. The training target was the price per square meter of each property (*price_per_sqm*), log-transformed as *log_price* for model fitting. A per-square-meter target was used because it places condominiums, houses, and bare lots on a comparable unit basis and reduces the distortion that raw total price introduces when property sizes differ widely. Predicted unit prices were multiplied by floor or lot area to recover a total price in Philippine Peso for reporting. The independent variables included structural, geospatial, and administrative features. Macroeconomic indicators were reviewed during study design but were not retained in the final fitted specification.

Rather than fitting one model across all property types, the study estimated three separate models, one for each property stratum: condominiums, houses, and vacant lots. This stratified design responded directly to a problem identified during the earlier defense: a single model trained on mixed property types must reconcile price-per-square-meter levels that differ several times over. Vertical condominium space, built houses, and bare land are priced on different logics, and pooling them inflates error and blurs interpretation. The hedonic literature supports this choice, finding that housing markets are segmented by the sorting of heterogeneous households and that stratified models fit better than pooled ones (Dröes et al., 2019; Usman et al., 2020). Splitting the data by property type let each model learn the value structure of its own market segment. The strata sizes are shown in Table 4.

Within each stratum, three supervised learning approaches were compared in order to balance interpretability and predictive performance, while also testing whether GIS-derived and administrative variables improved the model beyond structural features

Table 4

Open-Market Modeling Strata

Stratum	Rows	Scope note
Condominium	1,300	Vertical residential units
Houses	1,223	Single Detached, House and Lot, Townhouse
Vacant Lot	849	Bare land, area 80–2,000 sqm, price $\geq \frac{1}{2}$ BIR zonal
Total modeled	3,372	Rows entering the three strata

Note. The 3,372 modeled rows are drawn from the 3,616-row assembled open-market ABT. The 244-row difference comes from the model-requirement filters (no computable target, no same-type neighbor for the spatial-lag feature, or a vacant lot outside the scope band); Chapter 5 explains this filtering.

alone:

1. **Ordinary Least Squares (OLS) / Hedonic Regression**: An interpretable baseline grounded in Hedonic Pricing Theory (Rosen, 1974). Coefficients carried direct economic meaning (e.g., "each additional bedroom adds PHP X to value"). OLS was retained as a diagnostic comparator rather than the deployed model.

2. **Random Forest Regressor**: A tree-based ensemble that captured non-linear relationships and feature interactions without requiring explicit specification (Breiman, 2001). The Random Forest was the deployed model in all three strata.

3. **XGBoost Regressor**: A gradient boosting algorithm optimized for predictive accuracy on structured tabular data (T. Chen & Guestrin, 2016). It was evaluated under the same protocol as a second tree-based candidate (Nyanda et al., 2024).

Data Sources

Because direct deed-of-sale transaction records are not publicly accessible in the Philippines, the study used open-market online listings as the closest available proxy for current residential prices. Listings represent asking prices rather than settled transactions, but at scale they capture pricing patterns that are often missing from sparse official records (Sousa et al., 2024). To widen coverage and reduce dependence on any single platform's inventory, listings were collected from several property portals rather than one site. Table 5 summarizes the data sources and their roles.

Table 5

Summary of Data Sources Used in the Thesis Workflow

Source	Role	Volume	Nature
Open-Market Listings	Training label source	Lamudi, FilipinoHomes, DotProperty (3,616 rows)	Asking / market-facing
BIR Zonal Values	Administrative benchmark	Per barangay	Static / regulatory
Google Maps APIs	Geocoding and POI retrieval	Per property	Geospatial / dynamic
OpenStreetMap Road Network	Routing and network topology	Per property / destination pair	Geospatial / open data

Open-Market Listings

To represent current asking prices in the residential market, the study collected listings from public online property portals. These listings became the training scope for the deployed price-surface model, and they were gathered in two phases.

The first phase scraped Lamudi, the largest portal in the study, in two scraper generations. An initial collector used the *requests* library with HTML parsing for bulk retrieval. After Lamudi deployed a JavaScript challenge and CAPTCHA wall that blocked the bulk collector, a second browser-based scraper built on Playwright was added to recover the remaining listings. Together these yielded 1,578 listings from the bulk phase and 270 from the anti-bot browser batch.

The second phase widened coverage to two additional portals: FilipinoHomes, accessed through its backend JSON interface, which returned precise embedded coordinates for 1,203 listings; and DotProperty, which contributed 565 listings geocoded from their barangay-level address text. A fourth portal, OnePropertee, was scraped but dropped from the final dataset after its records were found to contaminate the model: its per-square-meter prices were frequently mis-extracted and its listings were geocoded to city centroids, which together produced large, systematic over-predictions on its rows. After merging, de-duplication, and the integrity filters described in Section 3.3, the open-market Analytics Base Table held 3,616 rows. A sample of the cleaned listing structure is documented in Appendix A.

Administrative and Macroeconomic Data

- **BIR Zonal Values**: Official zonal values per barangay, used as administrative benchmark features and as the basis for the derived valuation-gap diagnostic (Bureau of Internal Revenue, n.d.).

- **RPPI Review**: The BSP residential price index was reviewed during study design as a possible macro-control, but it was not retained in the final fitted specification (Bangko Sentral ng Pilipinas, 2025). The deployed model is cross-sectional and listing-level, so quarterly RPPI movements were not joined as an active model feature in the final ABT.

Geospatial Data Sources

- **Google Maps Geocoding API**: Converted property addresses into latitude/longitude coordinates and supported reverse geocoding of barangay names for the BIR join (Google LLC, 2025).
- **Google Maps Places API**: Supplied the raw point-of-interest inventories that were later curated into the MCRAI amenity categories.
- **OpenStreetMap Road Network**: Provided the street graph used for network-distance computation through the *osmnx* workflow (Boeing, 2017, 2019). In the final implementation, OSM was used for routing and network topology rather than as the primary POI source.

Target Variable

The predictive task of the thesis is the estimation of open-market residential price. The target variable was *price_per_sqm*, fitted on the log scale as *log_price*; predictions were back-transformed and multiplied by area to recover total price for reporting and for the OLS baseline comparison.

The complete feature matrix schema is documented in Appendix A.

Validation Layer: Human-in-the-Loop

Licensed real estate brokers from the CPRE network were included as a validation layer, not to replace statistical evaluation, but to check whether the model outputs remained plausible within local market practice. Their role was limited to two tasks:

1. **Sanity Check**: Reviewing SHAP-derived value driver rankings against practitioner knowledge.
2. **Outlier Review**: Examining cases with high prediction error to distinguish likely data issues from genuine market anomalies.

Data Pipeline

The data preparation process proceeded through the stages summarized in Figure 1 and described below, implemented in Python.

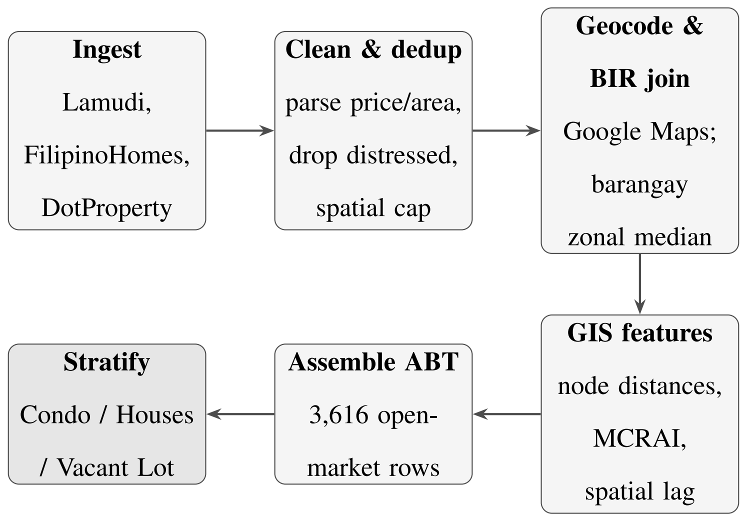


Figure 1

Data pipeline from multi-portal open-market ingestion to the modeling-ready Analytics Base Table and the three modeling strata.

1. **Ingestion**: Open-market listings were collected from Lamudi (bulk *requests* scraper plus a Playwright browser scraper), FilipinoHomes (backend JSON interface), and DotProperty (portal listings).

2. **Filtering**: The dataset was restricted to residential properties within the six study LGUs (Cebu City, Mandaue City, Lapu-Lapu City, Talisay City, Minglanilla, and Consolacion) using point-in-polygon checks against their boundaries.

3. **Parsing, Cleaning, and De-duplication**: Price and area fields were normalized across portals (currency symbols and commas stripped; rows without explicit numerical prices dropped). Distressed and mortgage-assumption listings (for example, “for assume” or “pasalo” postings, which quote loan balances rather than market value) were removed. Duplicate listings were collapsed on coordinate, price,

and area, and a spatial cap limited the number of listings per location cell. A per-stratum price-per-square-meter sanity band removed area-entry errors and extreme mispricing.

4. **Geocoding (Address to Coordinates)**: Property addresses were batch-geocoded through the Google Maps Geocoding API to obtain latitude and longitude. The service was selected for project-level address resolution in Metro Cebu, where listings often reference barangays, landmarks, or compound descriptors rather than standardized street-number formats. Results were cached locally to avoid redundant API calls. FilipinoHomes listings carried precise embedded coordinates and bypassed text geocoding.

5. **BIR Zonal Value Extraction and Barangay Join**: Official BIR zonal value schedules were sourced from the Revenue District Office files covering the six-LGU study area. A custom parser extracted street-level values by residential classification code and aggregated them to the barangay level as the median per classification. Each property was assigned a barangay through reverse geocoding, which served as the join key to the BIR summary table, yielding *bir_zonal_rr_median* coverage across the open-market ABT.

6. **GIS Augmentation**: From geocoded coordinates, the geospatial feature set described in Section 3.4.1 was computed: road-network distances to eight polycentric and regional anchor nodes; category-specific MCRAI component scores derived from curated amenity inventories; the *mcrai_composite*; and a spatial lag variable capturing the mean price of neighboring same-type properties within 500 m. Network shortest paths were computed on the OSM road graph, with Haversine fallback only when a point could not be snapped cleanly to the network.

7. **Final ABT Assembly and Stratum Split**: All engineered features were merged into a single Analytics Base Table and saved as a flat CSV. The modeling-ready

subset was produced by dropping rows with null *price_per_sqm* or *spatial_lag_price*, applying integrity flags, and then splitting into the condominium, house, and vacant-lot strata for separate model fitting.

Tools: Python (Pandas, NumPy, Scikit-learn, XGBoost, Statsmodels, Requests, Playwright, OSMnx), and the Google Maps Geocoding and Places APIs.

Feature Engineering

Engineered variables: *price_per_sqm*, *valuation_gap* (a diagnostic field, not a model feature), and derived log-price terms for the modeling target and baseline regression. The listing *source* field was kept in the ABT for auditing but excluded from the feature matrix, because it is not available at prediction time for a new property.

Geospatial Feature Engineering

Geospatial feature construction was the part of the method that most directly connected the valuation model to the urban structure of Metro Cebu. Rather than treating location as a background descriptor, this stage translated network accessibility, amenity access, and neighborhood context into variables that could enter the model explicitly.

1. **Geocoding and Administrative Join**: Property addresses were geocoded through the Google Maps Geocoding API to obtain coordinates, then reverse-geocoded to recover barangay names for the BIR zonal-value join described in Section 3.3. Results were cached locally to preserve API quota and ensure repeatable downstream feature construction.

2. **Network Distance Features**: For each property, shortest-path road distances were computed to eight polycentric and regional anchor nodes: Cebu Business Park, Mandaue CBD, Mactan CBD, South Road Properties, Talisay Tabunok, Consolacion, Naga City, and Mactan-Cebu International Airport. This node set followed the thesis' polycentric framing of Metro Cebu and retained both internal subcenters and major external anchors that shape residential accessibility gradients

Table 6

Feature Categories Used in the Thesis Workflow

Category	Features	Source
Structural	Lot Area, Floor Area, Bedrooms, Bathrooms, Property Type, imputation flags	Listing portals
Locational	City, Barangay, Latitude/Longitude	Google Maps API
Geospatial	Network distance to Cebu Business Park, Mandaue CBD, Mactan CBD, SRP, Talisay Tabunok, Consolacion, Naga City, and Airport	Geocoding + OSM road network
MCRAI	Eight component scores (education, grocery, health, hospitals, recreation, security, tourism, retail density) plus *mcrai_composite*	Places API POI inventory + network routing
Spatial Lag	Mean price of neighboring same-type properties within 500 m	Computed from data
Administrative	BIR zonal-value features per barangay	BIR schedules
Metadata	Source and market-segment fields retained in ABT but excluded from the fitted feature matrix	Engineered

(Giuliano & Small, 1991). These road-network distances also carry the transport-accessibility signal in the model, which is why transport was not retained as a separate amenity category. Distances were computed on an OpenStreetMap street graph using the OSMnx workflow (Boeing, 2017, 2019). When a property or destination could not be snapped cleanly to the network, a Haversine fallback was used so that valid observations were not dropped from the ABT.

3. **Metro Cebu Residential Accessibility Index (MCRAI)**: Amenity access was operationalized through the Metro Cebu Residential Accessibility Index (MCRAI), a custom gravity-style accessibility framework adapted to local valuation practice. Category-specific point inventories were assembled from Google Maps Places queries and cleaned into eight categories: education, grocery, health, hospitals, recreation, security, tourism, and retail density. Following Hansen-style accessibility logic, nearer opportunities contribute more strongly than farther opportunities (Hansen, 1959). For property i and category c, the category score was computed as:

$$MCRAI_{ic} = \sum_{j \in c} \frac{1}{\max(d_{ij}, 0.5)^{\beta}}, \qquad \beta = 2,$$

where d_{ij} is the road-network distance in kilometers from property i to amenity j, floored at 0.5 km to prevent singular values for immediately adjacent points. Each category was evaluated only within its designated search radius r_c, reflecting the fact that some amenities operate at neighborhood scale while others serve wider urban catchments.

The inverse-square form ($\beta = 2$) follows the conventional gravity-model setting, in which accessibility falls off with the square of distance, and the search radii in Table 7 were set to baseline catchment scales typical of each amenity type. Both are defensible starting values rather than parameters estimated from Metro Cebu data, so they are treated as a baseline in this study; Chapter 10 recommends deriving the

Table 7

MCRAI Categories and Search Radii

Category	Representative POIs	Radius (km)
Education	Schools, universities	2.5
Grocery	Supermarkets, wet markets	2.0
Health	Clinics, primary care	2.0
Hospitals	General and specialty hospitals	5.0
Recreation	Gyms, sports complexes, parks, beaches	1.5
Security	Police substations, fire stations, barangay halls	2.0
Tourism	Hotels, resorts, transient lodging	3.0
Retail Density	Restaurants, cafes, bakeries, convenience stores	1.0

decay rate and the radii empirically in future work.

The *mcrai_** component scores were available individually, and a separate *mcrai_composite* was retained as a summary index. The composite combined the positive-coefficient OLS categories—education, grocery, and recreation—with weights of 0.447, 0.345, and 0.222 respectively. Transport accessibility was moved to the road-distance features above, and finance was retired from the framework because no comparable Southeast Asian hedonic study treats banking proximity as a standalone residential amenity category. For categories that correlated negatively with price, the preferred interpretation is spatial sorting or threshold externality rather than direct residential amenity value generation (Bayer & McMillan, 2012; W. Y. Chen & Jim, 2010; Dronyk-Trosper, 2017; Song & Knaap, 2004; Tiebout, 1956; Yang et al., 2016).

4. **Spatial Lag Variable**: To capture neighborhood price effects, *spatial_lag_price* was computed as the mean price of other *same-type* properties within a 500 m neighborhood of each target property. Restricting the neighborhood to the same stratum kept the comparison like-for-like, and the 500 m radius matched the micro-scale at which the MCRAI amenity signals operate. Properties with no same-type neighbor within this radius received a null lag and were excluded from model fitting. This variable operationalized Tobler's First Law of Geography in a form that could be used directly by the regression and tree models (Tobler, 1970).

Pre-processing

Modeling Strategy

Stratified Models and Per-Stratum Feature Selection

Each property stratum was fitted separately, and the feature set for each stratum was trimmed using evidence from the exploratory analysis rather than kept identical across all three. The trimming drew on variance inflation factors, OLS coefficient significance, leave-one-block ablation, and MCRAI zero-rate checks. This produced a methodological contribution of its own: the model recognizes that different property types are priced by different geospatial structure.

All three strata dropped the redundant BIR log and commercial-zonal columns and the two road-class distance features, which added little once the eight node distances were present. The condominium and house models then collapsed the eight individual MCRAI categories into the single *mcrai_composite*, because for built homes the categories were strongly inter-correlated (roughly 0.57 to 0.96) and behaved as one accessibility summary. The vacant-lot model kept six individual MCRAI categories (education, grocery, health, hospitals, recreation, tourism), which were informative for bare land, but dropped the composite (which was collinear with its own components), security (an uneven data layer

Table 8

Pre-processing Steps Applied to the Modeling Subset

Step	Method	Rationale
Integrity Filtering	Remove rows with null *price_per_sqm*, null *spatial_lag_price*, or explicit integrity flags	Keeps the modeling subset consistent with the final pipeline
Log Transformation	Apply log transform to *price_per_sqm* to form the modeling target	Reduces right skew while keeping a per-unit target
Missing Values	Median imputation for remaining numeric fields; explicit zeros for structurally absent or all-null fields	Preserves usable rows while handling sparse engineered fields
Encoding	Dummy encoding for city and property type	Required for the fitted design matrix while keeping barangay for joins and reporting

with a high zero rate), and retail density (near-zero correlation with lot price). The net feature counts were 21 for condominiums, 24 for houses, and 22 for vacant lots, at accuracy comparable to the fuller feature sets. The plain reading is that built homes are summarized well by one accessibility index, while bare land responds to specific kinds of amenity access.

Hedonic Equation

The hedonic regression took the following log-linear form for the baseline comparator within each stratum:

$$\begin{aligned}\ln(\text{price\_per\_sqm}_i) = \alpha &+ \beta_1 \ln(\text{area}_i) + \beta_2(\text{bedrooms}_i) + \beta_3(\text{bathrooms}_i) \\ &+ \beta_4(\text{distance-to-node features}_i) + \beta_5(\text{MCRAI features}_i) \\ &+ \beta_6(\text{BIR zonal features}_i) + \beta_7(\text{spatial lag}_i) + \epsilon_i\end{aligned}$$

The dependent variable is the log of price per square meter (*log_price*), consistent with the training target used for OLS, Random Forest, and XGBoost. Predicted unit prices are back-transformed and multiplied by area to recover total price before the business-facing figures are reported. In this form, the hedonic model retains its interpretive structure while allowing administrative and spatial variables to enter directly into the price equation. This makes it possible to compare a more conventional valuation framework with the tree-based models without stripping out the locational features that are central to the study.

Evaluation Protocol: Leak-Free Cross-Validation

Model quality was estimated with five-fold grouped cross-validation (GroupKFold), where the groups were coordinate clusters. Listings that shared the same location were forced into the same fold so that a property used in training could not reappear, in near-identical spatial form, in the test fold. This guarded against the coordinate leakage that would otherwise flatter a spatial model, since the spatial lag and neighborhood features can make co-located listings look almost like copies of one another. All reported headline metrics were computed out-of-fold under this protocol.

Hyperparameter Tuning

For the tree-based models, a parameter grid was searched under the same grouped cross-validation, with the median absolute percentage error as the selection criterion. The Random Forest grid covered the number of trees, the share of features considered at each split, the minimum samples per leaf, and the maximum depth; the XGBoost grid covered the number of trees, learning rate, depth, and subsample. The per-stratum best settings were recorded in the deployment manifest. The tuning curves were close to flat across the search range, which indicated that the models were not very sensitive to these settings at this data size, and the deployed parameters sat at or near the defaults.

Evaluation and Explainability

Performance Metrics

The headline metrics were the median absolute percentage error (MdAPE) and the share of predictions within 20 percent of the actual price (PE20), because they describe how the model does for a typical property and resist distortion from a small number of extreme misses. Mean absolute percentage error (MAPE), the coefficient of dispersion (COD), and the price-related differential (PRD) were retained as supporting diagnostics, with COD and PRD reported in the spirit of assessment ratio studies. The study does not claim compliance with IAAO ratio-study standards; those thresholds are used only as context.

Benchmark Studies Used for Later Comparison

SHAP Explainability

SHAP was used to interpret model outputs at both the dataset and property level, which was important to keep the results reviewable in valuation practice and consistent with international valuation standards on transparency (International Valuation Standards

Table 9

Performance Metrics Used in Model Evaluation

Metric	Description	Purpose
MdAPE	Median Absolute Percentage Error	Headline; typical-property error, robust to outliers
PE20	Share of predictions within 20% of actual	Headline; plain-language accuracy rate
MAPE	Mean Absolute Percentage Error	Supporting; comparable across studies but tail-sensitive
COD	Coefficient of Dispersion	Supporting; uniformity of the ratio of predicted to actual
PRD	Price-Related Differential	Supporting; checks for regressivity (under/over-valuing by price level)

Council, 2025).

- **Global SHAP (Summary Plots)**: Identified which value drivers affected Metro Cebu property prices most across each stratum. This answered RQ1 ("What value drivers significantly influence property prices in Metro Cebu?").

- **Local SHAP (Force Plots)**: Explained individual predictions. For example: *"This Mactan condominium is predicted higher because of stronger neighborhood prices and accessibility, but the smaller floor area pulls the estimate down."*

This kept the model interpretable enough to work as a support tool rather than an opaque prediction system. The model was judged not just on how accurately it predicted price, but also on whether its outputs could be examined and discussed in terms practitioners understand.

Table 10

Benchmark Studies Used to Contextualize Later Evaluation Results

Study	Context	MAPE	Use in This Thesis
Ramolete et al. (2023)	Philippines, larger dataset	10–21%	Closest Philippine reference point; used for a like-for-like comparison under a matched split in Chapter 7.
Nyanda et al. (2024)	Tanzania, $n \approx 954$	48%	Small-sample emerging-market benchmark for the later evaluation chapter.

Deliverables

Web Decision-Support Application

The main applied output of the study was a prototype web-based decision-support application that organized model results across Metro Cebu. Rather than presenting results only as tables or summary statistics, the application was designed to show how predicted values, valuation-gap diagnostics, and locational patterns were distributed across the study area:

1. **Predicted Price Layers**: Point or surface outputs summarizing predicted open-market residential values across Metro Cebu.
2. **Valuation-Gap Diagnostics**: Comparison layers showing the divergence between model-based values and BIR zonal benchmarks.
3. **Supporting Locational Layers**: CBD-node, road-network, and

amenity-accessibility context used to interpret the price surface geographically.

The application used a TypeScript and Leaflet front end for the map interface, backed by a Python (FastAPI) service that ran the exact deployed Random Forest models for inference, so that no modeling logic was reimplemented on the client side. In practical use, the application centered on a predicted residential price surface for Metro Cebu while also supporting property-level prediction checks and SHAP-based explanation views. Its purpose was to make the model easier to inspect, both geographically, by comparing values across locations, and at the level of individual predictions and their underlying value drivers. Given the model's remaining error, the application is intended as a prototype for inspection and triangulation rather than a production valuation system. Appendix F (Pictures 1–3) shows its three views.

Data Understanding

This chapter covers the CRISP-DM data understanding phase for the frozen analytics base table (ABT). The goal here is simple: describe what the data looks like, where it is strong, and where it still has issues. This chapter does not discuss preprocessing or modeling steps. Those are covered in Chapter 5 and later chapters.

Initial Data Collection

The frozen open-market ABT had 3,616 rows and 51 columns, assembled from several online property portals into one file. This chapter describes that full assembled table. Not every row could serve as a training example: of the 3,616 rows, 3,372 met the modeling requirements and were used to fit the three stratified models. The remaining 244 rows are clean but lack a value the models require, either a computable target or the spatial-lag neighbor feature, and Chapter 5 details how they were filtered before fitting.

Table 11 traces each portal from raw scrape to the rows retained in the final ABT. Lamudi was the largest contributor, through a bulk scraper and a later browser-based batch that recovered listings after the site added an anti-bot wall. FilipinoHomes and DotProperty widened the coverage and reduced dependence on any single portal. A fourth portal, OnePropertee, was scraped but excluded entirely after its records were found to contaminate the model, so it appears here only as raw volume.

The ABT covered six LGUs: Cebu City, Mandaue City, Lapu-Lapu City, Talisay City, Minglanilla, and Consolacion. Naga City stayed as a CBD distance anchor but was not a training LGU. Cebu City and Lapu-Lapu City together held the largest share of rows, reflecting the concentration of online residential listings in the core and island submarkets.

Data Description (Schema)

The ABT columns can be grouped into eight blocks. The table below gives the main groups and their purpose.

Three columns are key for this thesis flow. First, *price_per_sqm* is the target variable. Second, *valuation_gap* is a diagnostic field based on *price_per_sqm* and the BIR

Table 11

Open-Market Collection Funnel and Composition by Source

Source	Raw scraped	Retained in ABT	Share
Lamudi (bulk)	4,477	1,578	43.6%
Lamudi (browser)	665	270	7.5%
FilipinoHomes	3,894	1,203	33.3%
DotProperty	3,721	565	15.6%
OnePropertee	3,804	—	—
Total	16,561	3,616	100.0%

Note. Retained counts are listings that survived geocoding, the six-LGU and residential filters, distressed-listing removal, the price-per-square-meter sanity band, deduplication, and the spatial cap. The browser-batch figure counts net-new listings after deduplication against the existing ABT. OnePropertee was excluded after its city-centroid geocoding and mis-extracted per-square-meter prices were found to contaminate the model.

residential benchmark. Third, the MCRAI block keeps both the eight category-level scores and the *mcrai_composite* score. The earlier finance category was retired and transport accessibility was moved to the road-distance features, so the framework now carries eight amenity categories.

This schema also shows why the ABT was not yet a final model matrix at this stage. Some fields were complete and stable, like CBD distances. Other fields depended heavily on source behavior, especially structural housing fields.

Initial Data Exploration

Initial exploration focused on target distribution, property-type mix, geographic spread, and accessibility coverage. The open-market *price_per_sqm* distribution was

Table 12

Analytics Base Table Schema by Feature Group

Feature Group	Representative Columns	Count	Role in Chapter 4 description
Identifiers and provenance	property_id, source, geocode_source, market_segment	6	Row identity and source provenance
Structural and flags	property_type, area_sqm, floor_area_sqm, lot_area_sqm, bedrooms, bathrooms, imputation flags	10	Physical attributes and structural sparsity
Location and scope	city, latitude, longitude, barangay_geocoded	4	Study-area placement
CBD-distance features	Eight dist variables for CBP, Mandaue, Mactan, SRP, Tabunok, Consolacion, Naga, and Airport	8	Network access to polycentric anchors
Administrative zonal fields	bir_zonal_rr_median, bir_zonal_cr_median, bir_zonal_rc_median, bir_zonal_rr_log	4	BIR benchmark linkage
Target and derived price fields	price_php, price_per_sqm, log_price, valuation_gap, price_outlier_flag	5	Target, transform, and diagnostics
Spatial context	spatial_lag_price	1	Neighbor-price context
MCRAI accessibility fields	Eight category scores plus mcrai_composite	9	Accessibility coverage summary
Total	Full ABT schema	51	Frozen chapter reference file

right-skewed, with a first quartile near PHP 47,500, a median near PHP 82,000, and a third quartile near PHP 152,000, with a long upper tail. This skew is why the modeling target was log-transformed.

Property-type composition is shown in Table 13. Condominiums were the largest class, and vacant lots had far lower median unit prices than vertical or built housing types. This mix is exactly why the study modeled the property types separately rather than pooling them.

Geographic spread in the open-market subset was uneven. Cebu City had the highest median unit price at about PHP 113,600 per square meter, followed by Mandaue

Table 13

Property-Type Composition and Median price_per_sqm (Open-Market ABT)

Property Type	Rows	Share	Median price_per_sqm
Condominium	1,391	38.5%	PHP 163,473
Vacant Lot	924	25.6%	PHP 29,106
House and Lot	856	23.7%	PHP 72,169
Single Detached	259	7.2%	PHP 80,000
Townhouse	184	5.1%	PHP 57,900
Apartment	2	0.1%	PHP 160,363

Counts are for the full open-market ABT before the per-stratum scope filters applied in Chapter 5.

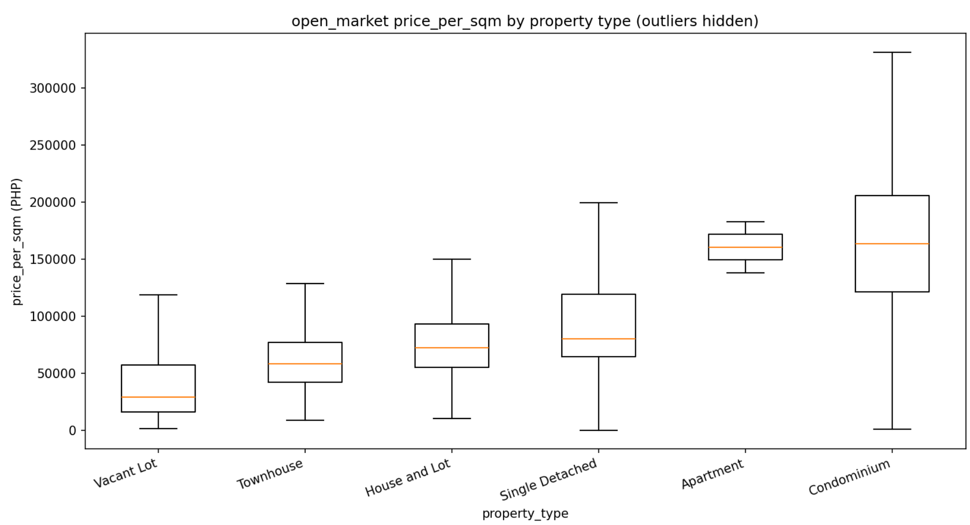
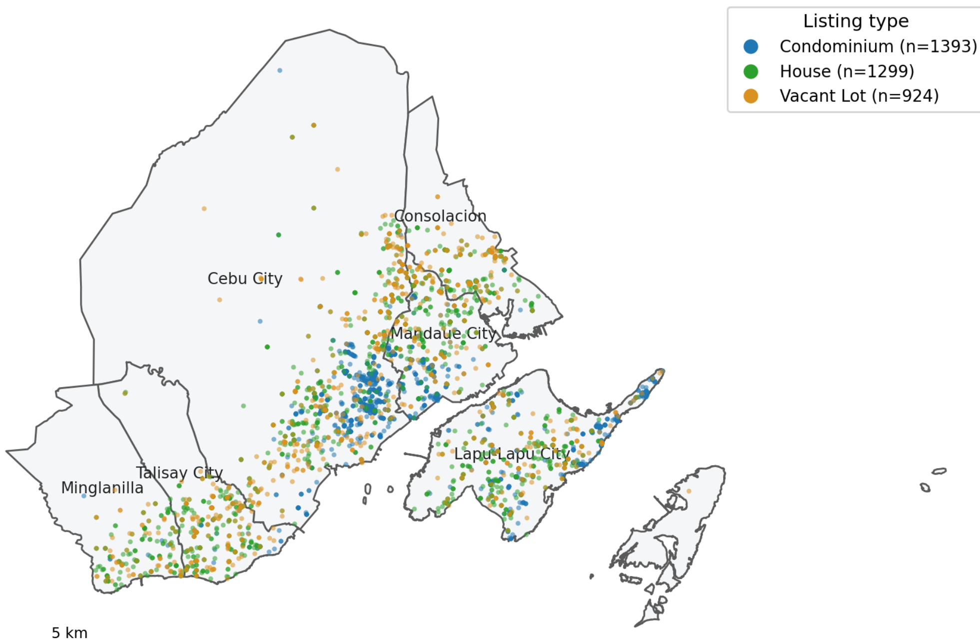


Figure 2

Distribution of price_per_sqm by property type (outliers hidden). Condominiums commanded the highest unit prices; vacant lots had the lowest, reflecting their land-only composition without built area.

City at about PHP 96,100 and Lapu-Lapu City at about PHP 92,100. Talisay City, Minglanilla, and Consolacion formed a lower-price band, with medians between about PHP 47,100 and PHP 56,800 per square meter. The premium open-market signal therefore stayed concentrated in the core and island submarkets rather than the peripheral suburban LGUs. The CBD-distance matrix also showed that several southern-corridor distances moved together closely, an early warning that the linear baseline would later need a collinearity check even though the nodes remained substantively meaningful. Map 2 shows how the listings were distributed across the study area by stratum.



Map 2

Open-market listings across Metro Cebu, colored by modeling stratum.

Table 14 shows that MCRAI coverage was generally strong, but not uniform. Education, the composite, tourism, and grocery had very low zero rates, while security and retail density had higher zero rates. The higher security zero rate, concentrated unevenly across LGUs, was one reason that category was later dropped from the vacant-lot model. Map 3 shows the spatial distribution of the curated amenity points

Table 14

Overall Zero Rates for Metro Cebu Residential Accessibility Index Fields

MCRAI Field	Zero Rate	Exploratory Reading
mcrai_security	26.6%	Security infrastructure was absent in many residential catchments
mcrai_retail_density	15.8%	Retail and food density improved after query expansion but was not universal
mcrai_recreation	12.0%	Recreation coverage was broad but thinner in some peripheral locations
mcrai_health	10.0%	Clinic access was uneven across the six-LGU scope
mcrai_grocery	4.3%	Grocery access was near-universal in the final file
mcrai_hospitals	3.1%	Hospital catchments were broad given the 5 km radius
mcrai_tourism	1.5%	Tourism and hospitality nodes were almost always observed
mcrai_education	0.3%	Education access was effectively complete in the final file
mcrai_composite	0.2%	Positive-weight composite coverage was effectively complete

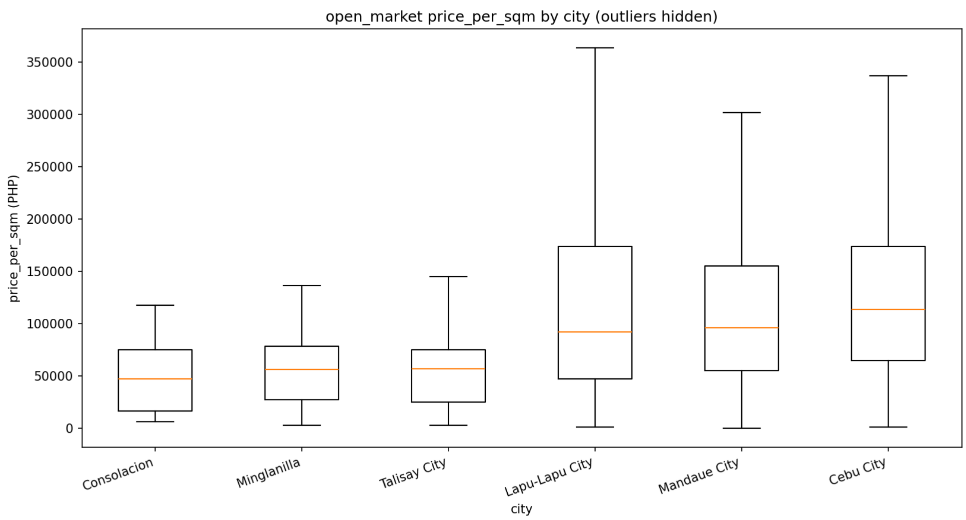
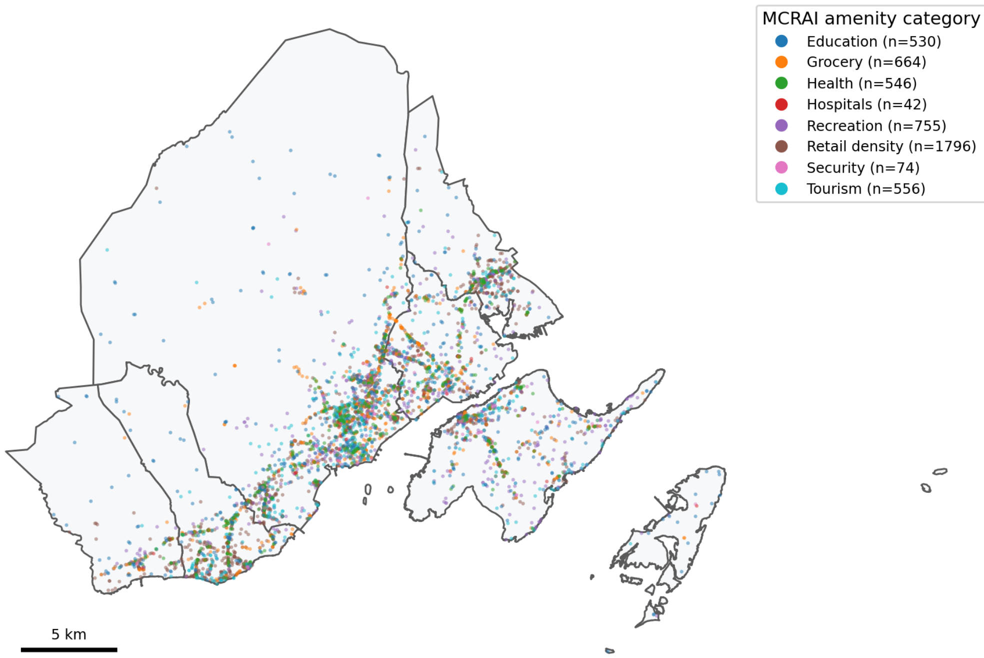


Figure 3

Distribution of price_per_sqm by LGU for the open-market subset (outliers hidden). Cebu City, Mandaue, and Lapu-Lapu showed the highest median unit prices; Talisay, Minglanilla, and Consolacion formed a lower-price band.

across the eight MCRAI categories.

A rank-based exploratory check on the open-market subset showed that location and accessibility fields aligned positively with unit price, while several size-related structural fields moved in the opposite direction. The BIR residential benchmark and the MCRAI composite were positively associated with *price_per_sqm*, while floor area, bedrooms, and bathrooms were negatively associated with it. This does not mean the target was inverted. It reflects the arithmetic of a per-square-meter target and the composition of the sample: compact condominiums in Cebu City and Lapu-Lapu City tend to command higher PHP per square meter, while larger detached homes and land-oriented listings more often sit in lower-density submarkets where total asking price can still be high but price per square meter is lower. Consistent with this reading, the same structural variables remained positively associated with total asking price.



Map 3

Curated amenity points across Metro Cebu for the eight MCRAI categories.

Data Quality Assessment

Data quality checks looked at missingness, integrity, source differences, and external-data contamination issues that were fixed before freeze.

The largest missingness issue was structural. Lot area was fully missing in the open-market file because listings report floor area rather than lot area; this reflects a source limitation, not random noise, and it is the reason the vacant-lot model relies on lot size while the other strata use floor area. The bedroom and bathroom gaps were concentrated in portal listings and were median-imputed with audit flags for the condominium and house strata, while vacant lots carried no bedroom or bathroom fields at all.

The ABT also kept soft outlier flags separate from hard integrity drops. The soft *price_outlier_flag* marked extreme unit prices without removing them automatically, while a smaller set of implausible rows was hard-dropped during cleaning. Source differences were also clear: combining several portals introduced source-level price

Table 15

Major Missingness Patterns in the Open-Market ABT

Field	Share Null	Reading for Chapter 5
lot_area_sqm	100.0%	Structural source absence; open-market listings carry floor area, not lot area
bir_zonal_rc_median	35.6%	Some residential-condominium zonal classes were missing by location
barangay_geocoded	22.4%	Location granularity remained uneven for vague addresses
bedrooms	18.2%	Mostly portal field gaps; imputed for condominiums and houses
bathrooms	17.5%	Mostly portal field gaps; imputed for condominiums and houses
bir_zonal_cr_median	9.1%	Some commercial zonal classes missing by location
spatial_lag_price	3.8%	Neighbor-based context unavailable for a small edge subset
price_per_sqm / valuation_gap	3.2%	Target derivation incomplete for a small subset
floor_area_sqm / area_sqm	2.5%	Residual portal incompleteness

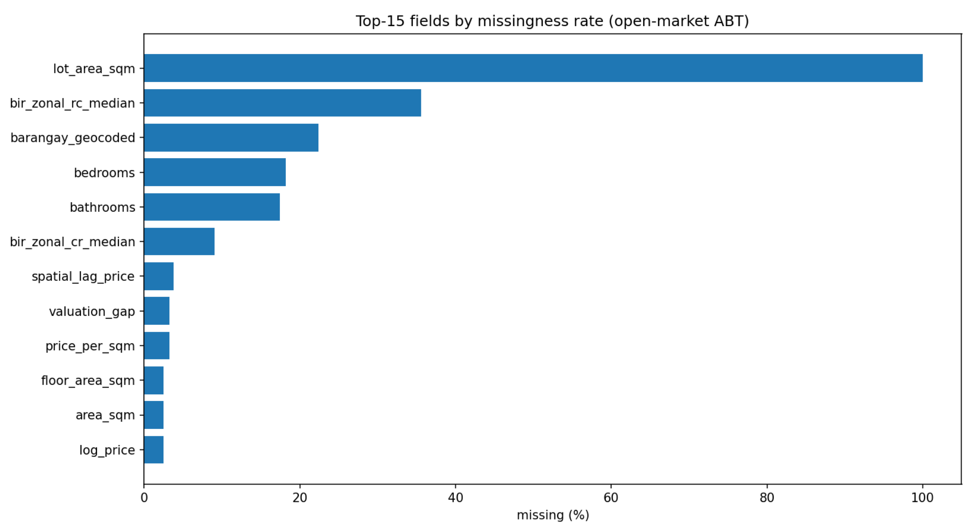


Figure 4

Top fields by missingness rate in the frozen open-market ABT. Lot area dominated because it was structurally absent in open-market listings rather than randomly missing.

effects, which is recorded as a limitation in Chapter 9 because the source field cannot be used at prediction time.

Another issue was fixed before freeze in the POI pipeline. Earlier runs had a bounding-box bug that allowed out-of-scope points to pass the LGU filter. The corrected freeze removed this contamination before final MCRAI recomputation.

Overall, the ABT at this stage was rich and usable for analysis, but still not model-ready in raw form. It had strong geospatial and benchmark coverage, but also source-driven structural gaps, a few implausible rows, and clear differences in price behavior across submarkets. Chapter 5 explains how these findings were handled in data preparation.

Data Preparation

This chapter covers the CRISP-DM data preparation phase for the frozen analytics base table (ABT). Chapter 4 described the condition of the data. This chapter explains what was done to turn that file into a modeling-ready dataset. The focus here is on filtering, missing-data treatment, feature engineering, encoding, and the stratum split used by the modeling script.

Scope of Preparation Work

The preparation work produced a frozen open-market ABT of 3,616 rows for the six-LGU study scope: Cebu City, Mandaue City, Lapu-Lapu City, Talisay City, Minglanilla, and Consolacion. Naga City stayed in the system only as a CBD distance anchor through *dist_naga_city_m*; it was not part of the training LGU set.

This open-market ABT was assembled from several listing portals: Lamudi (1,578 from the bulk scraper and 270 from the later browser-based anti-bot batch), FilipinoHomes (1,203), and DotProperty (565). A fourth portal, OnePropertee, was scraped but dropped after its records were found to contaminate the model with mis-extracted per-square-meter prices and centroid-level geocoding. The merge, de-duplication, distressed-listing removal, and price-band filters that produced the final 3,616-row ABT are described below, and Chapter 4 reports the resulting composition by source.

The goal of this phase was to turn that frozen ABT into the exact matrices used by the stratified modeling pipeline. This meant keeping the ABT as the reference file, then applying integrity filters and splitting the data by property type for separate model fitting.

Cleaning and Filtering

The cleaning that produced the open-market ABT followed the valuation basis of the thesis. Open-market listings represent the closest available proxy to Market Value in the IVS sense (International Valuation Standards Council, 2025).

Several filters were applied during the multi-portal cleaning step. Distressed and mortgage-assumption listings, which quote loan balances rather than market value, were removed by keyword. Duplicate listings were collapsed on coordinate, price, and area, and a spatial cap limited the number of listings per location cell so that a few dense clusters could not dominate a stratum. A per-stratum price-per-square-meter sanity band removed area-entry errors and extreme mispricing. One condominium record was dropped because it was a misclassified house-sized unit.

The remaining model-requirement filters were tied to the target and the spatial-lag feature. Rows with a null *price_per_sqm* were removed because the target could not be computed, and rows with a null *spatial_lag_price* were removed because the neighborhood-price feature was part of the final design. These filters reduced the assembled 3,616-row ABT to the 3,372-row modeling subset, drawn from the cleaned residential taxonomy: Condominium, House and Lot, Townhouse, Single Detached, and Vacant Lot.

Stratum Split

The modeling-ready data was split into three strata for separate model fitting, because condominiums, houses, and vacant lots are priced on very different per-square-meter logics. Table 16 shows the resulting sizes.

The vacant-lot stratum used an additional scope filter, keeping lots between 80 and 2,000 square meters with a price of at least half the BIR zonal value, to exclude commercial-scale parcels and obvious data errors.

Missing Data Treatment

The missing-data treatment kept rows usable while keeping every imputation visible and leaving structurally absent fields unimputed.

The first structural decision involved size variables. The thesis used *area_sqm* as the main size feature, with floor area preferred when available and lot area used as fallback. This mattered because vacant lots are priced on land area while condominium listings are priced on floor area. Vacant lots carried only this size field; they had no bedroom or bathroom attributes, and none were imputed for them.

Table 16

Open-Market ABT Split into Modeling Strata

Stratum	Rows	Membership
Condominium	1,300	Vertical residential units
Houses	1,223	Single Detached, House and Lot, Townhouse
Vacant Lot	849	Bare land within the lot scope filter
Total modeled	3,372	Rows entering the three strata

Note. The three strata sum to 3,372 of the 3,616 assembled open-market rows, and these 3,372 are the rows used to fit the models. The 244-row difference comes from the model-requirement filters applied at fitting: rows with no computable *price_per_sqm* target, rows with no same-type neighbor within 500 m for the spatial-lag feature, and vacant lots outside the lot-scope band (area outside 80–2,000 sqm or price below half the BIR zonal value). These rows are clean but cannot serve as training examples, so they remain in the assembled ABT for description while the stratified models are fit on the 3,372-row modeling subset.

For condominiums and houses, bedrooms and bathrooms were sometimes missing and were median-imputed, with audit flags retained so imputed values stayed visible. In the condominium stratum, 242 rows had an imputed bedroom value and 146 an imputed bathroom value; in the house stratum, 39 rows had an imputed bedroom value and 62 an imputed bathroom value. After filtering and encoding, any remaining numeric nulls were median-imputed as a final safeguard, and structurally absent columns were filled with zero rather than imputed.

Market Segment Treatment

The deployed task of the thesis was the estimation of open-market residential prices, and the dataset was made up of open-market listings only. The deployed map therefore estimates open-market residential price levels across Metro Cebu; it does not estimate forced-sale or administrative floor-price values.

Feature Engineering

Most of the heavy feature engineering had already been completed before this chapter, but Chapter 5 still had to decide which engineered fields would enter each stratum's matrix.

The project kept the eight network-distance variables: Cebu Business Park, Mandaue CBD, Mactan CBD, SRP, Talisay Tabunok, Consolacion, Naga City, and Airport distances. For accessibility, the ABT carried eight category-level Metro Cebu Residential Accessibility Index (MCRAI) fields plus the *mcrai_composite*. The older *amenity_score_** fields had already been dropped because they duplicated the same concept with a weaker formulation, and the retired finance category and transport category were removed, with transport accessibility represented through the road-network distances instead.

The *spatial_lag_price* field, the mean price of other same-type properties within 500 meters, was treated as a required engineered feature rather than an optional one. Rows with a null spatial lag were removed during filtering instead of being imputed.

As described in Chapters 3 and 6, the feature set was then trimmed per stratum from exploratory evidence. All three strata dropped the redundant BIR log and commercial-zonal columns and the two road-class distance features. The condominium and house models used the single MCRAI composite, while the vacant-lot model used six individual MCRAI categories and dropped the composite. The OLS baseline used a slightly tighter set within each stratum, replacing the raw area fields with a log-area term and removing variables that were collinear with adjacent CBD-distance terms, to keep the hedonic coefficients stable.

Encoding and Final Modeling-Ready Matrices

The final preparation step converted each stratum into model matrices. City and property type were one-hot encoded with *drop_first = True*. After encoding and numeric-null handling, the retained feature counts were 21 for condominiums, 24 for houses, and 22 for vacant lots, as summarized in Table 17. Instead of a single train-test split, model quality was estimated with five-fold grouped cross-validation by coordinate cluster, so that co-located listings could not leak across folds.

Table 17

Final Modeling-Ready Matrices by Stratum

Stratum	Rows	Features	Validation
Condominium	1,300	21	GroupKFold (5)
Houses	1,223	24	GroupKFold (5)
Vacant Lot	849	22	GroupKFold (5)

Overall, the data preparation phase was conservative. It applied a small number of defensible filters, preserved the main valuation signals, kept the matrices complete enough for model fitting, and split the data by property type. Chapter 6 discusses how these prepared matrices were used in OLS, Random Forest, and XGBoost.

Modeling

This chapter covers the CRISP-DM modeling phase for the prepared open-market dataset. Chapter 5 described how the training data was assembled. This chapter explains how the three models were specified, fit, and validated within each property stratum in the current pipeline. Detailed evaluation results are discussed in Chapter 7.

Model Selection Rationale

The study used three models because each one served a different role. Ordinary Least Squares (OLS) was the interpretable hedonic baseline rooted in implicit-price theory (Rosen, 1974). Random Forest was included because it can capture non-linear relationships and feature interactions without a fixed functional form (Breiman, 2001). XGBoost was included because gradient boosting usually performs well on structured tabular data (T. Chen & Guestrin, 2016). Chapter 3 introduced these models conceptually; the focus here is the fitting setup used in the saved implementation.

Stratified Fitting and Leak-Free Validation

Rather than fitting one model across all property types, the pipeline split the open-market data into three strata and fit a separate set of models for each: condominiums (1,300 rows), houses (1,223 rows), and vacant lots (849 rows). All models used the same target, *log_price*, the log of price per square meter, which kept the training target consistent across OLS, Random Forest, and XGBoost. Predictions were back-transformed and multiplied by area to recover total price for reporting.

Model quality was estimated with five-fold grouped cross-validation (GroupKFold), where the groups were coordinate clusters. Listings that shared a location were kept together in the same fold, so a property could not appear in both training and test folds in near-identical spatial form. This was a deliberate guard against coordinate leakage, which would otherwise flatter a spatial model whose neighborhood and spatial-lag features make co-located listings resemble copies of one another. All headline metrics in Chapter 7 were computed out-of-fold under this protocol. A separate analysis re-ran the models under a conventional random 80:20 split to compare against an external benchmark study; that comparison is reported in Chapter 7 and is not the basis for the deployed metrics.

Table 18

Stratified Modeling Setup

Item	Value	Notes
Strata	3	Condominium, Houses, Vacant Lot
Condominium rows	1,300	Fitted separately
House rows	1,223	Fitted separately
Vacant-lot rows	849	Fitted separately
Validation	GroupKFold (5)	Groups = coordinate clusters (leak-free)
Random seed	42	Reproducible fitting and folds
Common supervised target	log_price	Log of price per square meter

Per-Stratum Feature Sets

The feature matrix was not identical across strata. As described in Chapter 3, each stratum's features were trimmed from evidence (variance inflation factors, OLS significance, leave-one-block ablation, and MCRAI zero rates). All three strata dropped the redundant BIR log and commercial-zonal columns and the two road-class distance features. The condominium and house models then collapsed the eight individual MCRAI categories into the single *mcrai_composite*, while the vacant-lot model kept six individual

MCRAI categories but dropped the composite, security, and retail density. The resulting feature counts were 21 for condominiums, 24 for houses, and 22 for vacant lots.

The tree-based models used each stratum's full retained feature set. The OLS baseline used a slightly tighter version within each stratum, removing variables that were highly collinear with adjacent CBD-distance terms and replacing raw area fields with a log-area term, so the baseline followed a stable log-linear hedonic form. These tighter controls were a stability choice for coefficient interpretation, not a claim that the dropped columns lacked predictive value, since tree splits tolerate redundant and correlated predictors more easily.

OLS Hedonic Baseline

Within each stratum the OLS baseline was fit in *statsmodels* on the log price-per-square-meter target. It was the most constrained specification of the three because it had to remain stable enough for coefficient interpretation. Heteroscedasticity and non-normal residuals were treated as diagnostic issues for the OLS comparator and addressed with robust (HC3) standard errors; they were not blockers for the deployed tree models. OLS was retained as an interpretable comparator and as the basis for screening the MCRAI category signs, not as the deployed model.

Random Forest

The Random Forest was the deployed model in all three strata, fit with *scikit-learn*'s *RandomForestRegressor* using 300 trees (Breiman, 2001). The per-stratum settings selected under grouped cross-validation are listed in Table 19. No scaling step was applied before fitting, since tree models are invariant to monotone feature scaling.

XGBoost

The XGBoost comparator was fit with the *XGBRegressor* implementation from the *xgboost* package (T. Chen & Guestrin, 2016), on the same per-stratum feature sets and the same *log_price* target as the Random Forest. This kept the comparison fair across the two tree-based models. As Chapter 7 reports, XGBoost performed comparably to the Random

Table 19

Deployed Random Forest Settings by Stratum

Stratum	Trees	Max features	Min samples / leaf
Condominium	300	0.7	1
Houses	300	1.0	2
Vacant Lot	300	1.0	1

Maximum depth was left unbounded in all three strata.

Forest in every stratum but did not surpass it.

Hyperparameter Tuning

The tree-based models were tuned with the same grid search and grouped cross-validation set out in Chapter 3, using the median absolute percentage error as the selection criterion so that tuning stayed aligned with the headline deployment metric. The tuning curves were close to flat across the search range, which indicated that the models were not very sensitive to these settings at this data size, and the deployed parameters sat at or near the defaults. The per-stratum hyperparameter sweep figures are included in the appendix material in `EDA/plots/11_hyperparameter_tuning/`.

MCRAI Composite Weight Derivation

The MCRAI composite was built from the OLS coefficient signs rather than by forcing all categories into one additive score. It retained only the positive-coefficient household-access categories and renormalized them into a three-part weight vector: 0.447 for education, 0.345 for grocery, and 0.222 for recreation. Transport accessibility was represented separately through the road-network distance features rather than as a composite category, and finance was retired from the framework.

The remaining categories, including security, tourism, and retail density, were not used in the composite because their OLS coefficients were negative or near zero, which is interpreted in Chapter 8 as spatial sorting rather than direct disamenity. The condominium and house models used the composite as their single accessibility summary, while the vacant-lot model used the individual categories instead, because for bare land the categories were not interchangeable.

Table 20

MCRAI Composite Weights (Positive-Coefficient Categories)

Category	Weight	Role in the Current Workflow
mcrai_education	0.447	Composite (condo/house) and standalone feature (lot)
mcrai_grocery	0.345	Composite (condo/house) and standalone feature (lot)
mcrai_recreation	0.222	Composite (condo/house); not retained for lots

To avoid rank deficiency, the composite and its component categories were not used together in the same model: the condominium and house models used the composite, and the vacant-lot model used the individual components.

SHAP Setup

SHAP was configured after model fitting for the deployed Random Forest in each stratum using *shap.TreeExplainer* (Lundberg & Lee, 2017). The explainer was applied to each stratum's data, mean absolute SHAP values were computed by feature, and per-stratum beeswarm summaries were exported. Because the OLS baseline already had an interpretable coefficient structure, SHAP was not needed for that part of the pipeline. The SHAP step served as the technical bridge between model fitting and the explainability analysis discussed in Chapters 7 and 8.

Evaluation

This chapter presents the CRISP-DM evaluation phase for the final set of fitted models. Chapter 6 explained how the OLS baseline, Random Forest, and XGBoost were specified and trained within each property stratum. The focus here is the leak-free cross-validated evidence used to judge model quality, select the production model, and examine the error pattern of the retained model. Broader interpretation of what these results mean for Metro Cebu valuation practice is reserved for Chapter 8.

Evaluation Approach

All reported metrics were computed out-of-fold under five-fold grouped cross-validation, with coordinate clusters as the groups, so that co-located listings could not appear in both training and test folds. Predictions were made on the log price-per-square-meter target, then back-transformed; percentage errors were computed on the per-square-meter scale.

The study led with two headline metrics and kept three more as supporting diagnostics. The median absolute percentage error (MdAPE) measured the error for a typical property and was chosen as the headline because, unlike the mean, it is not dragged around by a small number of extreme misses. The proportion of predictions within 20 percent of the actual price (PE20) gave a plain-language accuracy rate that a practitioner could read directly.

The mean absolute percentage error (MAPE) was retained as a supporting metric because it is the figure most other studies report, which makes cross-study comparison possible, though it is sensitive to the error tail. The coefficient of dispersion (COD) measured how tightly the ratio of predicted to actual price clustered, and the price-related differential (PRD) checked whether the model valued cheaper and more expensive properties even-handedly. COD and PRD are reported in the spirit of assessment ratio studies; the thesis does not claim compliance with IAAO ratio-study thresholds, and those values are used only as context.

Final Model Comparison

Table 21 presents the head-to-head comparison of the three approaches within each stratum. The direction was the same across property types: the two tree-based models beat the OLS hedonic baseline in every stratum, by about two percentage points of MdAPE for houses and by five to six points for condominiums and vacant lots. The Random Forest edged out XGBoost in all three strata by a small margin. The gap between Random Forest and XGBoost was narrow enough to call the two comparable, with Random Forest the modest winner.

Table 21

Leak-Free Cross-Validated Comparison by Stratum (Median Absolute Percentage Error)

Stratum	OLS	Random Forest	XGBoost	PE20 (RF)	PRD (RF)
Condominium	24.5	**19.3**	19.8	51%	1.21
Houses	25.1	**22.7**	23.6	44%	1.20
Vacant Lot	44.8	**38.4**	40.2	26%	1.48

MdAPE values per model; PE20 and PRD shown for the deployed Random Forest. Supporting diagnostics for the Random Forest: condominium COD 37.7 / MAPE 36.5; houses COD 35.8 / MAPE 35.1; vacant lot COD 55.9 / MAPE 58.3.

The OLS baseline trailed the tree models in every stratum, which confirmed that a linear hedonic specification was useful as an interpretable comparator but was not the best predictive option for this data. The condominium and house models reached a typical error near 19 and 23 percent, while the vacant-lot model was clearly the hardest, with a typical error near 38 percent. That ordering is revisited in the residual analysis below.

Best Model Selection

The Random Forest was selected as the production model in all three strata. It posted the lowest MdAPE of the three approaches in every stratum, and it did so while remaining the simpler tree model to deploy and maintain, since it depends only on scikit-learn and is less sensitive to parameter choices than gradient boosting at this data size. Because the Random Forest advantage over XGBoost was small, the deployment decision is framed honestly as “Random Forest best or tied, deployed for robustness and simplicity” rather than as a decisive accuracy win.

The comparison also showed a clear division of roles. OLS remained the interpretable hedonic benchmark used for coefficient-based reference and for grounding the MCRAI composite weights. The tree models were the stronger predictors. The evaluation phase therefore retained the Random Forest as the production estimator for each stratum while keeping OLS as the econometric baseline.

Do the Geospatial Features Help? (Ablation)

To test whether the engineered geospatial features earned their place, each stratum’s Random Forest was refitted in three tiers under the same folds: structural features only; then adding administrative location (city and BIR zonal value); then adding the full geospatial set (node distances, MCRAI, spatial lag). Table 22 reports the change in typical error.

Table 22

Ablation by Feature Tier (MdAPE, Leak-Free Cross-Validation)

Stratum	Structural	+ Administrative	+ Geospatial
Condominium	24.9	23.0	**19.3**
Houses	27.1	22.3	**23.0**
Vacant Lot	51.5	42.3	**38.6**

The + Geospatial tier adds the full geospatial set (all node distances, all MCRAI categories, and the spatial lag). The deployed models use the per-stratum trimmed feature sets described in Chapter 6, so their MdAPE differs slightly from this column (Table 21).

Compared with a structural-only model, the full geospatial set improved every stratum: condominiums by about 5.7 points, houses by about 4.2 points, and vacant lots by about 13.0 points. The decomposition is more revealing. When the engineered geospatial features were added *on top of* administrative location, they carried a clear additional gain for condominiums (about 3.7 points) and vacant lots (about 3.8 points), but added essentially nothing for houses (about -0.7 points, a slight worsening). For houses, knowing the city and the BIR zonal value already captured most of the location signal, and the custom geospatial features mostly duplicated it. The reading is that the model's engineered features add the most value precisely where administrative benchmarks are weakest, which is vertical condominiums and bare lots.

SHAP Feature Attribution

SHAP was applied to each deployed Random Forest so that its prediction logic could be described transparently. Because the models are stratified, the value-driver structure differs by property type, and the per-stratum SHAP importance in Figures 5–7 shows this directly. These charts rank each feature by its mean absolute SHAP value, that is, the average size of its effect on the predicted price. The directional beeswarm versions, which also show whether high or low feature values push the prediction up or down, are provided in Appendix C. Grouping the SHAP contributions into blocks gives the clearest summary:

- **Condominium**: neighborhood price level (spatial lag, about 34 percent of the SHAP magnitude) and CBD-distance access (about 30 percent) led, with unit size and bedrooms next. Administrative benchmarks contributed little.
- **Houses**: distance to the Cebu Business Park was the single strongest feature, and CBD distances as a block (about 39 percent) led, followed by structural size (about 35 percent) and neighborhood price.
- **Vacant Lot**: distance to the Cebu Business Park alone accounted for close to half of the model's attribution, and the individual MCRAI amenity categories together

carried about 22 percent, their largest share in any stratum.

Across all three strata, location-related features (CBD distances, MCRAI, and the spatial lag) carried the large majority of the attribution, which is consistent with the value-driver question in RQ1. The substantive interpretation of why these features dominate in Metro Cebu is developed in Chapter 8.

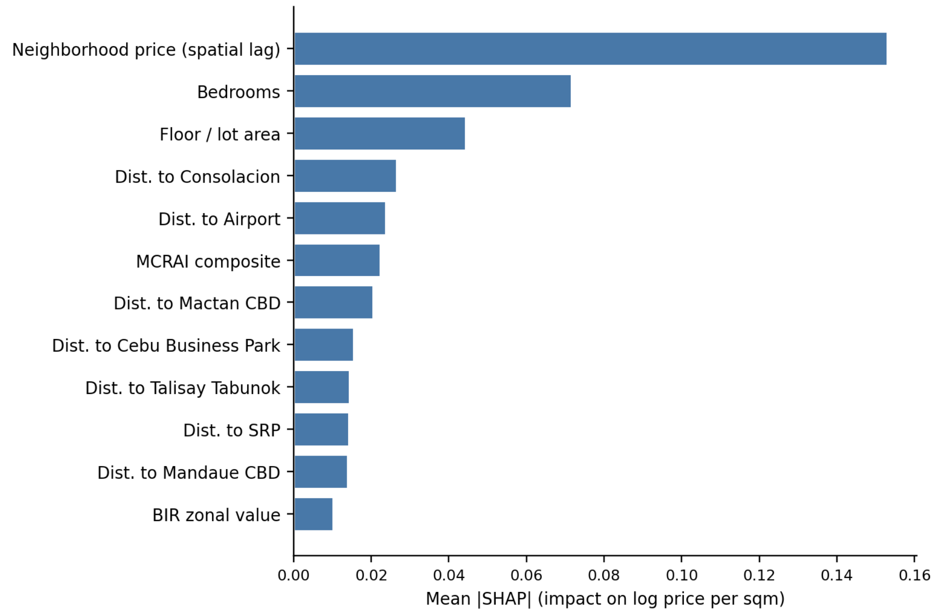


Figure 5

Mean absolute SHAP value by feature, deployed condominium Random Forest (top 12 features).

Residual Analysis

The residual pattern was consistent and informative across all three strata. At the center of the distribution the models were close to unbiased, with median signed errors within a few percent of zero. The damage was concentrated in the error tail. For each stratum, the worst tenth of predictions were properties priced far below the norm for their location and size, and the model predicted the location-implied price and overshot: the worst-decile condominiums were over-predicted by about 106%, houses by about 97%, and vacant lots by about 181%.

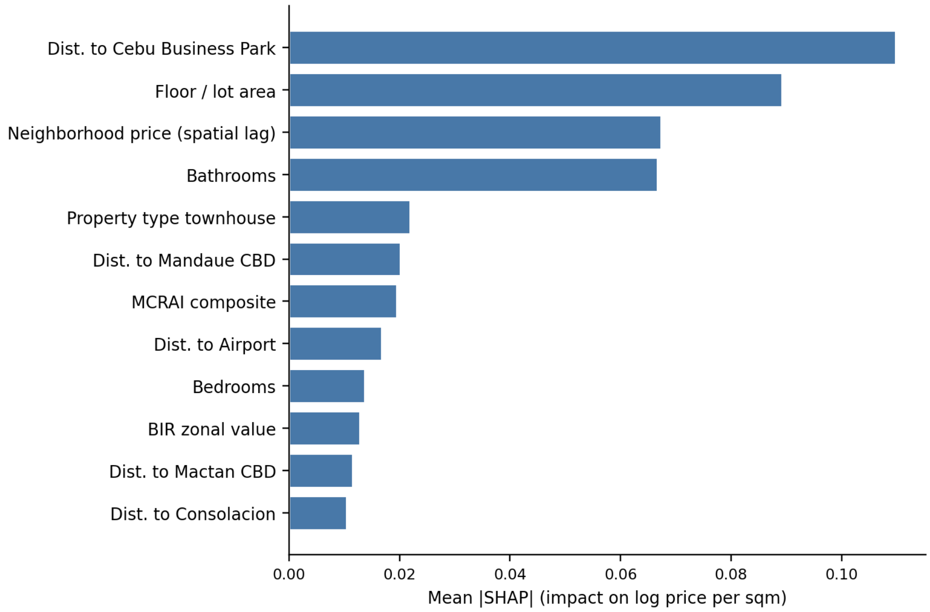


Figure 6

Mean absolute SHAP value by feature, deployed house Random Forest (top 12 features).

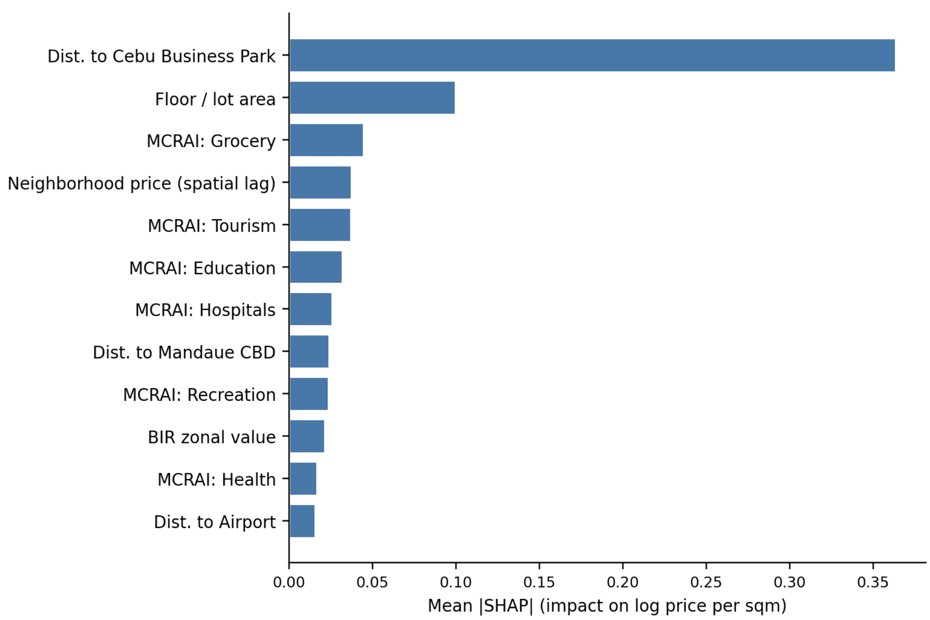


Figure 7

Mean absolute SHAP value by feature, deployed vacant-lot Random Forest (top 12 features).

These extreme misses were not random. They were atypically cheap listings whose low price reflects something the listing does not record, such as condition, title issues, distressed or pre-selling pricing, or, for lots, terrain and access. The model sees a normal location and size and reasonably predicts a normal price. This is the same finding seen three ways: the PRD above 1.0 in every stratum (a mild tendency to over-value cheaper properties), the gap between the good typical error and the larger average error, and the heavy worst-decile over-prediction. The vacant-lot stratum showed the strongest version of this, which is why its typical error stayed near 38 percent. Chapter 9 records the specific data limitations behind this tail.

Benchmark Comparison

The closest Philippine reference point is Ramolete et al. (2023), who reported MAPE in the 10 to 21 percent range using a richer feature set on a larger, house-dominated dataset under a random train-test split. Two adjustments make the comparison fair. First, the present models were re-evaluated under their random 80/20 protocol rather than the stricter grouped cross-validation, which raised the Random Forest MAPE only modestly (about 5.1 points for condominiums, 1.3 for houses, and 1.8 for vacant lots). This showed that coordinate leakage was a small part of the gap, not the main story. Second, leading with the median rather than the mean, the typical condominium error of about 16 percent and house error of about 22 percent under the matched split sit at the top of Ramolete's band, so the median property is competitive while a hard-case tail inflates the mean.

The remaining gap is genuine and explainable rather than a sign of weaker method. Ramolete's house sample was much larger (about 3,212 houses versus 1,223 here), the Cebu market is thinner, and their model used socio-economic features from PSA and DTI sources that this study did not include. The fairest like-for-like comparison is the house stratum, since their data is house-dominated.

Two further benchmarks frame the result. Nyanda et al. (2024) reported a boosting MAPE near 48 percent and a Random Forest near 53 percent in a noisy, mixed-market

Tanzanian setting, which is the kind of emerging-market, listing-based context most similar to this one. Against tightly controlled automated valuation systems, by contrast, the present errors are clearly higher; commercial and assessment-grade systems report single-digit median errors on cleaner transaction data. The honest reading is that this model beats the local baselines that matter for the thesis, the OLS hedonic and the BIR zonal benchmark, under an honest evaluation, but it is not assessment- or transaction-grade. Sousa et al. (2024) support the underlying choice: large online listing datasets reveal price segmentation that sparse official records miss, at the cost of a noisier target carrying asking-price spread, seller strategy, and submarket mixing.

For that reason the result is described plainly. The deployed Random Forest is the strongest model in the workflow under leak-free evaluation, and its typical-property error is competitive with the closest Philippine study, but it still carries a real error tail that the thesis acknowledges rather than hides.

Results and Discussion

Chapter 7 established which model performed best within each property stratum. This chapter shifts to interpretation. The point is not to repeat the evaluation tables, but to explain what the retained findings mean for residential valuation in Metro Cebu. Read as a whole, the results point to a market that is polycentric, strongly segmented by property type, and only partly captured by simple counts of nearby amenities.

Overview of Findings

Three points stand out. First, location dominated value formation in every stratum: the geospatial block of features (distances to the polycentric nodes, MCRAI amenity access, and the neighborhood price lag) carried the large majority of the SHAP attribution for condominiums, houses, and vacant lots alike. Second, the value-driver structure differed by property type, which is why the study modeled the three strata separately rather than together. Neighborhood price level led for condominiums, distance to the Cebu Business Park led for houses and lots, and the individual amenity categories mattered far more for bare land than for built homes. Third, the MCRAI results were selective rather than uniform. Education, grocery, and recreation carried positive signs and fed the composite, while security, tourism, and retail density behaved as diagnostic categories rather than additive premiums.

Taken together, these findings suggest that Metro Cebu's residential price surface is not organized around one center with a smooth decline outward. It behaves more like a corridor with several active peaks and overlapping catchments. That reading is consistent with the broader polycentric literature and with the local planning view of Metro Cebu as a system of linked urban clusters rather than a single dominant core (Giuliano & Small, 1991; Heikkila et al., 1989; Japan International Cooperation Agency & Metro Cebu Development and Coordination Board, 2015; McMillen, 2003).

Polycentric Structure in the SHAP Attributions

The per-stratum SHAP results are the clearest evidence that the model learned a polycentric rather than monocentric price structure. Distance to the Cebu Business Park was the single strongest feature for both houses and vacant lots, confirming that the historic core still anchors the market. But it did not stand alone. For condominiums, the neighborhood price lag led, followed by a spread of node distances rather than one: distance to Consolacion, to the airport, to the Mactan CBD, and to the Cebu Business Park all carried weight. This is the signature of a market that responds to several active nodes at once.

This does not mean the traditional core disappeared. For houses and lots, the Cebu Business Park gradient was the dominant locational signal, which fits the bid-rent logic in which land value falls with distance from the principal center (Alonso, 1964; Muth, 1969). The fuller picture is layered: the old core still matters most for land and detached housing, while the condominium market spreads its locational weight across the airport-linked Mactan corridor and the northern Mandaue and Consolacion nodes. The result fits the polycentric logic of Giuliano and Small (1991), the uneven gradient argument of McMillen (2003), and the question raised by Heikkila et al. (1989) about whether subcenters weaken a single-CBD explanation. In Metro Cebu, the market appears layered rather than centerless.

That interpretation also agrees with the JICA roadmap, which treats Metro Cebu as a linked urban system with specialized clusters and corridor growth nodes such as SRP and Naga (Japan International Cooperation Agency & Metro Cebu Development and Coordination Board, 2015). The continued response to nodes beyond the old core is consistent with the roadmap's treatment of the metropolis as a corridor of linked centers rather than a single hub.

Property-Type Segmentation and the Mactan Submarket

Property segmentation is part of the main structure of value formation, not a side result. In this study it is built into the method itself: the three strata are modeled separately because the same location does not generate the same premium across condominiums, detached houses, and vacant lots. The clearest demonstration is the difference in what drives each model. Condominium value leans on neighborhood price level and a spread of accessibility nodes; house and lot value lean on the bid-rent gradient to the Cebu Business Park; and bare land alone gives the amenity categories a large role. A single pooled model would have averaged these distinct logics into one blurred equation.

This matters for the Mactan side of the market. The results do not support an exaggerated claim of a raw island premium. Instead, the Mactan signal comes through the condominium stratum, where distance to the Mactan CBD and to the airport are among the stronger locational features. The premium is therefore better read as a submarket effect than as an island dummy. What the market appears to price is the bundle of airport access, bridge-linked connectivity, tourism-adjacent development, and condominium-oriented demand that characterizes the Mactan corridor.

That reading is consistent with Agosto's (2020) finding that mobility and livability matter for Cebu land values. It also fits the JICA roadmap's view of the tri-city core as functionally linked but internally differentiated (Japan International Cooperation Agency & Metro Cebu Development and Coordination Board, 2015). A useful regional comparison is the Bangkok condominium study, which found that transport-related and domestic accessibility factors can explain a large share of condominium value formation in a congested Southeast Asian market (Tochaiwat et al., 2023).

MCRAI Findings and the Spatial Sorting Interpretation

The MCRAI results need a more careful reading than a simple positive-versus-negative split. The final composite retained education, grocery, and recreation because those categories carried positive Stage 1 OLS coefficients in the open-market sample. That result is sensible because these variables describe everyday residential convenience and lower routine household friction. Transport accessibility was represented separately through the road-network distance features rather than as an amenity category, and finance was retired from the framework.

A second, sharper finding came from the stratified design: the amenity categories mattered very differently across property types. For condominiums and houses, the individual MCRAI categories were strongly inter-correlated and were summarized well by the single composite, contributing only a small share of attribution. For vacant lots, the individual categories together were the second-largest driver after the Cebu Business Park distance. The plain reading is that built homes are valued through one bundled accessibility summary, while bare land responds to specific kinds of nearby amenity access. This difference is itself a result, and it justified keeping individual MCRAI categories for lots while collapsing them to the composite for the other two strata.

The negative signs on security, tourism, and retail density should not be read as proof that those categories are inherently harmful. A better interpretation is spatial sorting. In a hedonic framework, households sort across locations according to the bundles of attributes they prefer and can afford (Bayer & McMillan, 2012; Tiebout, 1956). A variable can therefore carry a negative coefficient not because the amenity itself is undesirable, but because it is concentrated in places with congestion, visitor pressure, or commercial intensity that push some residential buyers away. The broader capitalization literature supports this conditional reading (Brasington & Parent, 2024; W. Y. Chen & Jim, 2010; Dronyk-Trosper, 2017; Song & Knaap, 2004; Yang et al., 2016). That also lines up with the Metro Manila hedonic report of Tiglao and Rivera (2006) and the Bangkok condominium study (Tochaiwat et al., 2023), both of which show that the market rewards specific forms of accessibility rather than all forms equally.

Comparison to Prior Cebu and Manila Hedonic Literature

The findings are broadly consistent with the earlier Cebu literature, but they refine it. Agosto's (2020) conclusion that mobility and livability matter is still visible in the present model. The road-network distances dominate value across strata, and the positive amenity categories mostly describe the daily convenience side of livability. What the current model adds is a stronger spatial reading: this capitalization does not operate evenly across the metropolis, but through several active nodes and property-specific submarkets.

The Metro Manila comparison points in the same direction. Tiglao and Rivera (2006) treated accessibility to work and urban opportunities as central to both land valuation and location choice. The Metro Cebu results support that idea, but they also suggest that accessibility in a polycentric region is not a single measure. It is a set of overlapping access logics: access to the old core, access to corridor nodes, access to airport-linked condominium markets, and access to daily neighborhood services. In that sense, Metro Cebu sits between a simple monocentric model and the stronger polycentric claim associated with Heikkila et al. (1989).

A regional comparison also helps explain why the MCRAI categories did not all move in the same direction. The Bangkok condominium article reported that transport accessibility and related factors could account for a large share of total model weight in that market (Tochaiwat et al., 2023). The present study found the same broad importance of accessibility, but in a more selective way, because the Metro Cebu model covered several residential property types and a wider mix of corridor conditions.

Model Trade-offs

The Random Forest was the best predictive choice within each stratum, but the findings show why model selection in this thesis could not be treated as a pure one-metric contest. OLS remained useful because it exposed coefficient direction, supported the MCRAI category screening, and showed that the Cebu Business Park still kept a meaningful role in the price structure. The tree-based models were needed because the market contains non-linear interactions across distance, corridor position, and property type that a single linear equation cannot represent well. The Random Forest and XGBoost performed comparably, and the Random Forest was deployed for its robustness on small per-stratum samples and its simpler implementation, rather than because of a decisive accuracy gap.

This is consistent with recent housing-prediction work that treats tree ensembles as strong tabular learners while still warning that data quality and market heterogeneity remain the real constraints (Nyanda et al., 2024; Sousa et al., 2024).

Limitations

Several limits should stay visible when reading these findings, and Chapter 9 records them in full. The model was trained on open-market listings rather than transaction records, so the target variable reflects asking-price behavior as well as market-value expectations. The vacant-lot stratum was the hardest to predict because bare-land value depends on parcel attributes that listings rarely record. The MCRAI categories also summarize accessibility through carefully designed but still simplified proxies; they cannot fully represent route quality, travel-time variability, perceived safety, or institutional quality.

There is also a methodological limit in the way spatial heterogeneity was handled. The tree-based models captured non-linearity and interaction effects, but the thesis did not estimate spatially varying coefficients directly. That matters in a polycentric region because the value of distance to a node is unlikely to be constant across all locations (Heikkila et al., 1989; McMillen, 2003).

Even with these limits, the main interpretation is stable. Metro Cebu's residential valuation pattern is best read as a polycentric market shaped by multiple active centers, property-type segmentation, and selective capitalization of accessibility. Chapter 9 builds on this interpretation by drawing the main conclusions of the thesis, while Chapter 10 turns those conclusions into the study's formal recommendations.

Conclusions

Summary of the Study

This study examined how residential property values in Metro Cebu could be estimated more consistently using a data-driven workflow with geospatial feature engineering. The problem began with the gap between fast-moving market prices and slower administrative or appraisal benchmarks. To address that problem, the study assembled a cleaned analytical base table from open-market listings across several online portals, BIR zonal values, and geospatial layers derived from Google Maps and OpenStreetMap. The deployed prediction scope covered open-market residential property in six LGUs: Cebu City, Mandaue City, Lapu-Lapu City, Talisay City, Minglanilla, and Consolacion.

The study followed the CRISP-DM process and compared three modeling approaches within each property stratum: OLS hedonic regression, Random Forest, and XGBoost. Rather than fitting one model across mixed property types, it estimated separate models for condominiums, houses, and vacant lots, on a per-square-meter target. The workflow also constructed a Metro Cebu Residential Accessibility Index (MCRAI) using category-specific radii and gravity-style decay, then paired that index with polycentric distance features, structural property variables, and a neighborhood price lag. Under leak-free grouped cross-validation, the deployed Random Forest reached a typical error (median absolute percentage error) of about 19 percent for condominiums, 23 percent for houses, and 38 percent for vacant lots. The thesis therefore concluded that a machine-learning workflow grounded in local urban structure can produce a usable decision-support tool for Metro Cebu open-market residential valuation, even if prediction error remains substantial for some property types and locations.

Answers to the Research Questions

Research Question 1 asked, “What value drivers significantly influence property prices in Metro Cebu?” The study found that location dominated value formation in every stratum, with the geospatial block of features carrying the large majority of the SHAP attribution. The specific driver differed by property type: neighborhood price level led for condominiums, while distance to the Cebu Business Park led for houses and vacant lots, the latter showing the clearest bid-rent gradient. The individual amenity categories mattered most for bare land. The conclusion was that property prices in Metro Cebu were shaped by a combination of polycentric accessibility and housing-submarket segmentation, not by one uniform citywide gradient.

Research Question 2 asked which modeling technique among Hedonic Regression, Random Forest, and XGBoost is most suitable for deployment. Under leak-free grouped cross-validation, the two tree-based models beat the OLS hedonic baseline in every stratum, by about two percentage points of typical error for houses and by five to six points for condominiums and vacant lots, and the Random Forest edged out XGBoost by a small margin throughout (condominium 19.3 versus 19.8, houses 22.7 versus 23.6, vacant lot 38.4 versus 40.2). Because the gap between the two tree models was narrow, the conclusion is framed as “Random Forest best or tied, deployed for robustness and simplicity” rather than as a decisive accuracy win. The Random Forest was retained as the production model in all three strata, with OLS kept as an interpretable diagnostic baseline.

Research Question 3 asked whether geospatial features significantly improve model performance compared with structural-only models. A three-tier ablation under the same folds answered this directly. Adding the full geospatial set to a structural-only model improved every stratum: condominiums by about 5.7 points, houses by about 4.2 points, and vacant lots by about 13.0 points of typical error. The decomposition was more revealing: when the engineered geospatial features were added on top of administrative location (city and BIR zonal value), they carried a clear extra gain for condominiums (about 3.7 points) and vacant lots (about 3.8 points), but added essentially nothing for houses (about -0.7 points). The conclusion was that geospatial features materially improved the model, and that they added the most value precisely where administrative benchmarks were weakest, namely vertical condominiums and bare lots.

Research Question 4 asked how large the valuation gap is between the model's data-driven predictions and traditional BIR zonal values. The study answered this both as a measurable field and as a quantified result. Using the clean land-to-land comparison for vacant lots, market price per square meter ran about three times the BIR zonal benchmark overall, ranging from about 2.2 times in Cebu City to about 4.8 times in Mandaue City. Across the study area, listings exceeded the BIR benchmark in close to 100 percent of cases in every LGU except Cebu City lots. The condominium and house percentage gaps are not directly comparable, because they divide a floor-area unit price by a land-area benchmark, so the defensible headline is the vacant-lot comparison. The conclusion was that BIR zonal values sit well below open-market levels across Metro Cebu, and that the gap is large, positive, and systematic rather than incidental.

Methodological Contribution

The methodological contribution of the thesis rests on the way established valuation ideas were adapted to the specific conditions of Metro Cebu. First, the study modeled property types separately and selected each stratum's features from evidence rather than using one identical set. This produced a finding of its own: different property types are priced by different geospatial structure, so built homes were summarized well by a single accessibility composite while bare land needed individual amenity categories. Second, the thesis built a polycentric distance feature set instead of relying on a single central-business-district measure, and a two-stage MCRAI workflow that used OLS signs to screen categories and kept only the positive household-access categories in the composite. Finally, all headline results were estimated under leak-free grouped cross-validation, which guarded against the coordinate leakage that would otherwise flatter a spatial model. Together, these decisions produced a locally grounded valuation workflow that was more specific to Metro Cebu than a generic hedonic template, and evaluated more honestly.

Practical Contribution

The practical contribution of the thesis is the decision-support layer that came out of the modeling work. The main deliverable is a prototype web-based application that presents a predicted open-market residential price surface for Metro Cebu, with property-level prediction checks and SHAP-based explanations. It used a TypeScript and Leaflet front end backed by a Python service that ran the exact deployed models. Because the model still carries substantial error for some property types, especially vacant lots, the application is presented as a prototype for triangulation rather than a tool for confident stand-alone valuation. It does not replace formal appraisal judgment, but it gives brokers, appraisers, and researchers a reproducible spatial estimate of residential market price under a clearly defined market segment.

The thesis also organized the model outputs spatially through the web application, including the valuation-gap diagnostic. This supports map-based review of predicted price levels and of where official benchmarks and market-facing estimates diverge, without treating the benchmark itself as the prediction target.

Limitations

Several limitations should stay visible when reading these conclusions.

- **Listing-based target.** The model was trained on open-market asking prices rather than recorded transaction prices, so the target carries asking-price spread, seller strategy, and submarket mixing.

- **Source heterogeneity.** Combining several portals introduced source-level price effects. FilipinoHomes listings, for example, were priced lower at equal features by roughly 14 percent, and the source field cannot be used at prediction time, so this remains a residual source of error.

- **Vacant-lot data ceiling.** The vacant-lot stratum was the hardest (about 38 percent typical error) because bare-land value depends on parcel attributes that listings rarely record, such as frontage, zoning, title status, slope, and flood exposure. The honest figure is a data ceiling, not a modeling failure; an earlier, more optimistic figure came from a smaller and geographically concentrated sample.

- **Condominium pre-selling contamination.** The cheap tail of condominium listings is partly contaminated by pre-selling down-payments or reservation fees scraped as if they were full unit prices, mostly in the new-development LGUs. This is a measurement error in the target; it was documented rather than corrected, because distressed-listing detection is weak when listing descriptions are not retained.
- **Intra-city terrain.** The model does not see elevation or terrain within a city. The largest vacant-lot over-predictions were genuinely cheap mountain or highland barangays inside Cebu City, which the model values using city-wide land levels.
- **Centroid-snapped geocoding.** A share of houses and lots with vague addresses were geocoded to barangay or subdivision centroids, which adds spatial-feature noise on those rows, although it does not break the leak-free cross-validation.
- **No spatially varying coefficients.** The tree models captured non-linearity and interactions, but the study did not estimate coefficients that vary across space, which a polycentric region would justify as future work.

Closing Statement

The thesis concluded that data science methods can support residential valuation in Metro Cebu when they are applied with clear market definitions, local geospatial logic, and disciplined treatment of value concepts. The strongest results did not come from model complexity alone. They came from matching the workflow to the structure of the local market: an open-market target, property-type segmentation, a polycentric urban form, and an accessibility framework designed for household valuation rather than for liquidation or administrative pricing. Under those conditions, the study produced a defensible decision-support tool for Metro Cebu residential valuation.

Recommendations

Recommendations for Valuation Practice

The price surface works best as a triangulation tool rather than a replacement for existing references. The deployed Random Forest was the strongest predictive option in the study, but Chapter 7 showed that the held-out error stayed substantial in some locations and property types, well above the accuracy expected of assessment-grade valuation systems. Practitioners should therefore read the surface alongside BIR zonal values and property-specific comparable evidence instead of treating the model output as a stand-alone final appraisal. When weighing those comparables, the variables that carried the most predictive weight deserve the most attention: Chapters 7 and 8 showed that polycentric distance and property type mattered more consistently than the accessibility composite, so corridor position, proximity to active subcenters, and housing submarket class should come before broad amenity counts.

The spatial sorting result from the Stage 1 OLS model needs careful interpretation. The negative coefficients on security, tourism, and retail density did not mean these uses are always harmful to residential value. They pointed to context effects such as congestion, commercial intensity, or neighborhood sorting. Practitioners should avoid assuming that physical proximity to police infrastructure, tourism strips, or dense commercial activity automatically adds residential value without checking the surrounding land-use context.

The valuation gap between market listings and BIR zonal values should be read as a research signal, not as an appraisal input. The thesis quantified the gap and preserved it as a diagnostic field: for the clean land-to-land comparison on vacant lots, market prices ran roughly three times the BIR zonal benchmark, and listings exceeded the benchmark in close to all LGUs. This indicates that official benchmarks lag the open market across Metro Cebu. It does not, however, establish a correction factor that can be applied during an appraisal. The gap is computed from asking prices through a model that is not assessment-grade, so it points to where benchmarks may be stale and warrants closer study, rather than supplying a value adjustment that practitioners can use directly. Validating the gap against recorded transactions is a prerequisite before any operational use.

Recommendations for Policy

The findings support continued polycentric planning in Metro Cebu. Chapters 7 and 8 showed that the market did not behave as if Cebu Business Park alone structured residential value; distance signals from Mactan, Consolacion, and Naga also mattered strongly. For regional planners and local governments, investment outside the historical core should keep recognizing the functional role of secondary and corridor-based subcenters rather than concentrating policy attention on the CBP corridor alone.

MCRAI may be useful as a reusable template for parcel-level accessibility, but as a structure to adapt rather than a finished index to copy. The thesis built MCRAI for residential valuation, not for nationwide equity mapping and not for liquidation pricing. Its category-specific radii, locally defined accessibility categories, and positive-only composite rule give a workable starting point for LGU-level planning or assessment work. Its parameters, however, were set on judgment-based grounds rather than estimated from local data, so they should be refined along the lines set out below before the framework is relied on for planning decisions.

Recommendations for Future Research

The most important next step is to put the geospatial accessibility features on a more objective and empirical footing. The current Metro Cebu Residential Accessibility Index rests on several design choices that were defensible but largely a matter of judgment rather than estimates from local evidence: the gravity decay parameter was fixed at $\beta = 2.0$, the search radii were set by category, and the composite weights were read from OLS coefficient signs. The framework also adapts an established accessibility-index logic rather than one calibrated specifically to Metro Cebu. Because the MCRAI categories contributed only modestly for condominiums and houses, and because the model's accuracy was limited overall, these arbitrary choices are a real candidate explanation that future work should test directly. A more objective approach would estimate the decay rate and the category weights from the data, compare alternative accessibility formulations against the current one, and treat known negative-effect amenities such as gasoline stations and dense tourism strips as nonlinear or threshold features rather than folding them into a single positive-only score. The aim is a spatial feature set whose construction is reproducible and data-driven rather than dependent on parameter values chosen in advance.

Spatial heterogeneity also deserves more direct estimation. The present models captured non-linearity but did not estimate coefficients that vary across space. Geographically Weighted Regression or Mixed Geographically Weighted Regression is the logical extension, and a follow-up study should test whether the capitalization of Cebu Business Park, Mactan, Consolacion, SRP, and Naga changes materially across submarkets instead of assuming one average distance effect for the whole region.

Broadening the model's inputs is the other major direction, since this study found that geospatial features and open-market listings are not sufficient on their own to value every property type well. Future work should collect richer parcel and structural attributes, such as road frontage and a corner-lot flag, which the vacant-lot data ceiling indicated are priced by the market but are rarely recorded in listings. It should also strengthen the time dimension of the analytical base table by preserving collection dates and adding macroeconomic indicators, such as exchange rates, inflation, and interest rates, so that market cycles and price drift can be modeled directly rather than left out of a cross-sectional design (Udomsap & Abid, 2020). A further extension is to triangulate predictions against complementary reference prices, such as BIR zonal values and bank reserve prices, instead of relying on a single market-facing target.

A Note on Deployment

Deployment was not a primary aim of this study, so only a few practical notes apply to the prototype web application. It should be run inside the project virtual environment so that its dependencies, including SHAP, load correctly, and it should display its open-market-only scope clearly so the price surface is not read as a universal market value. If the prototype is maintained, the point-of-interest inventory, listing coverage, and benchmark schedules will drift over time, so the data and predicted layers should be refreshed and re-verified periodically.

Closing Note

This thesis should be read as a demonstrated workflow more than as a finished valuation product. Its main contribution is that the process is reproducible: a custom MCRAI design, a polycentric distance feature set, and a Random Forest plus SHAP pipeline can be rebuilt, checked, and adapted for other Philippine secondary cities with similar data conditions. What this study finished was a defensible template for local valuation analytics, not the last version that Metro Cebu will ever need.

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Appendix A

Data Documentation

Figure A1 shows a slice of the assembled analytics base table as stored, illustrating the row-level structure behind the schema tables that follow. The full table holds 51 columns; the slice shows a representative subset.

id	source	city	type	area_sqm	beds	baths	price_php	price/sqm	bir_zonal	mcrai_comp	spatial_lag
119	Lamudi	Lapu-Lapu City	Condominium	71.0	1	1	21,900,000	308,451	7,000	8.7	36,210,629
120	Lamudi	Cebu City	Condominium	105.0	3	3	18,300,000	174,286	30,000	76.0	17,534,923
121	Lamudi	Lapu-Lapu City	Condominium	45.0	1	1	9,500,000	211,111	5,000	3.4	10,812,318
123	Lamudi	Lapu-Lapu City	Condominium	25.0	1	1	4,468,775	178,751	10,000	9.3	5,056,686
124	Lamudi	Lapu-Lapu City	Condominium	68.0	1	1	23,900,000	351,471	7,000	8.7	32,838,280
125	Lamudi	Consolacion	Single Detached	354.0	4	4	28,000,000	79,096	4,500	29.1	19,434,044
126	Lamudi	Lapu-Lapu City	Condominium	30.0	1	1	3,562,600	118,753	5,000	13.3	7,694,560
128	Lamudi	Lapu-Lapu City	Condominium	37.0	1	1	6,000,000	162,162	5,000	22.0	8,458,711
129	Lamudi	Cebu City	Single Detached	566.0	4	4	115,000,000	203,180	8,500	2.8	

Figure A1

Snapshot of the assembled open-market ABT (12 of 51 columns, first rows).

Table A1

Sample Structure of Cleaned Open-Market Listing Records

Raw Field (Web Scrape)	Processed Feature	Description
Location	Barangay, City	Parsed location string
Price	Actual Price	Target variable (PHP), cleaned of currency symbols
Bedrooms	Bedrooms	Numeric extract
Bathrooms	Bathrooms	Numeric extract
Floor area	Floor Area	Numeric extract (sqm)
Land Size	Lot Area	Numeric extract (sqm)
Latitude / Longitude	Latitude, Longitude	Direct coordinate mapping

Table A2

Final Feature Matrix Schema Used for the Modeling Workflow

Feature Group	Variables	Type
Identifiers / Metadata	Property ID, source portal (stored in ABT; excluded from the final fitted feature matrix)	Metadata
Structural	Lot Area, Floor Area, Bedrooms, Bathrooms, Property Type, vacancy / imputation flags	Numeric / Categorical
Locational	City, Barangay, Latitude, Longitude, *is_mactan_island*	Numeric / Categorical
Geospatial	Network distance to Cebu Business Park, Mandaue CBD, Mactan CBD, SRP, Talisay Tabunok, Consolacion, Naga City, and Airport	Numeric (meters)
MCRAI	*mcrai_education* through *mcrai_retail_density*, plus *mcrai_composite*	Numeric (index scores)
Administrative / Neighborhood	BIR zonal-value features and *spatial_lag_price*	Numeric (PHP)
Target	*price_per_sqm*, *price_php*, *log_price*	Numeric

Appendix B

Supplementary Tables

Table B1

Geographic Distribution of the Final Open-Market ABT by LGU and Property Type

City or Municipality	Rows	Share	Condo	House	Lot
Cebu City	1,368	37.83%	669	360	339
Lapu-Lapu City	870	24.06%	444	273	153
Mandaue City	548	15.15%	241	182	125
Talisay City	372	10.29%	19	214	139
Consolacion	242	6.69%	8	131	103
Minglanilla	216	5.97%	10	141	65
Total	3,616	100.00%	1,391	1,301	924

Note. The Condo, House, and Lot columns group the full assembled ABT by property type; the House column aggregates house-and-lot, single-detached, townhouse, and apartment listings. These full-ABT groups (1,391, 1,301, and 924) are larger than the modeling strata (1,300, 1,223, and 849) because 244 rows were held out of fitting by the model-requirement filters described in Chapter 5. Chapter 4 reports the finer property-type breakdown.

Appendix C

SHAP Beeswarm Plots

Chapter 7 reports SHAP feature importance as mean absolute values (bar charts). The beeswarm plots below carry the same ranking but add the directional detail. Each dot is one property; its horizontal position is that property's SHAP value (how much the feature pushed the predicted price per square meter up or down), and its color is the underlying feature value (red high, blue low). A band of red dots on the right therefore means high values of that feature raise the prediction. The vertical spread only separates overlapping dots and has no other meaning.

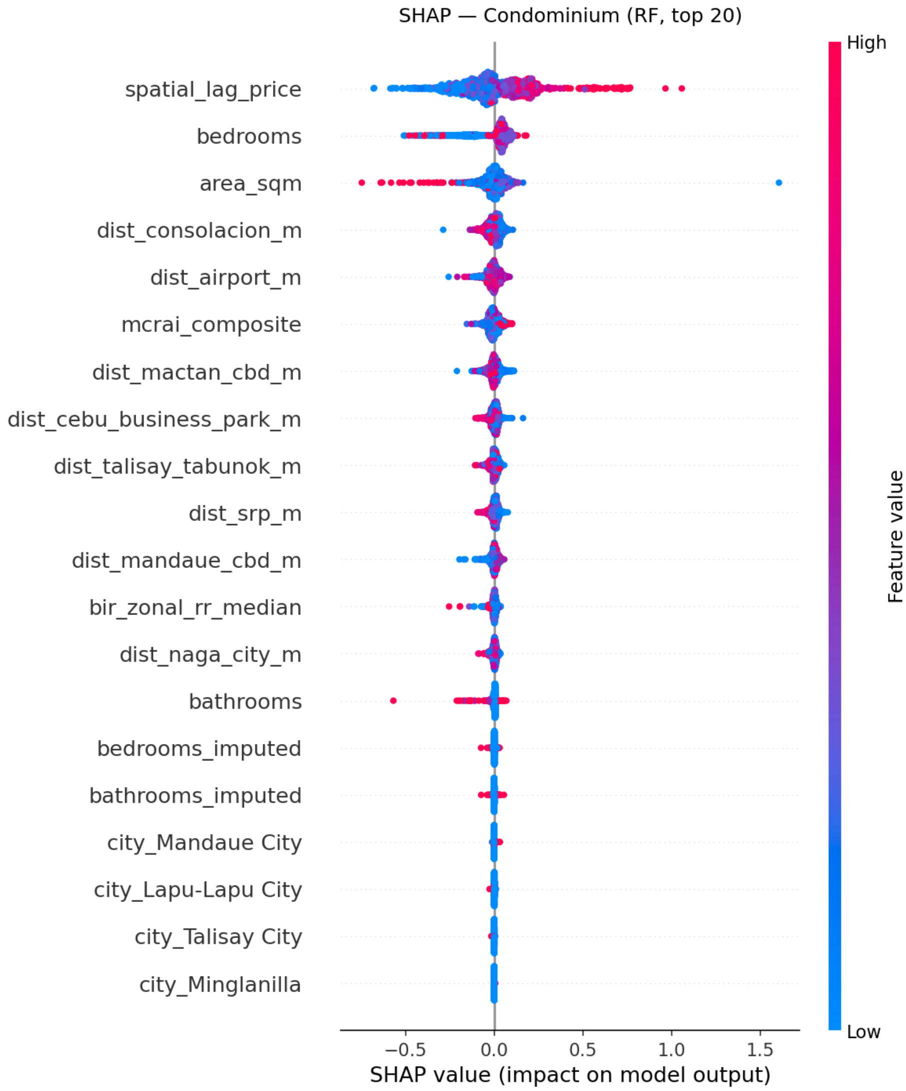


Figure C1

SHAP beeswarm summary, deployed condominium Random Forest.

Figure C2

SHAP beeswarm summary, deployed house Random Forest.

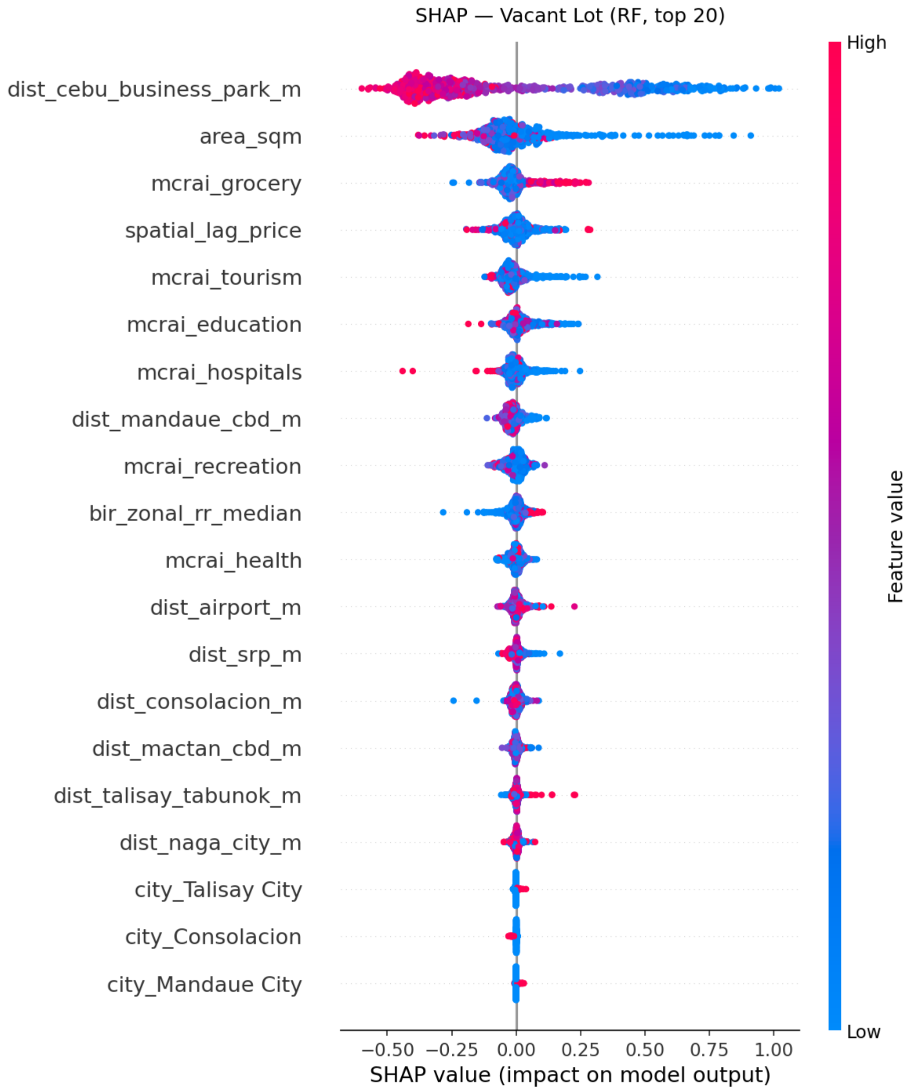


Figure C3

SHAP beeswarm summary, deployed vacant-lot Random Forest.

Appendix D

Feature Snapshots by Stratum

The tables below show example properties from each modeling stratum together with their full model-feature values, illustrating the per-stratum feature sets described in Chapters 3 and 6. Each column is one open-market listing chosen near the 30th, 50th, and 70th percentile of price per square meter, so the examples are representative rather than extreme. The one-hot city indicators use Cebu City as the reference category, so a property with zeros in all city rows is in Cebu City.

Table D1

Feature snapshot — Condominium stratum (21 model features).

Feature	Property 1	Property 2	Property 3
Floor / lot area (sqm)	22	34	69
Bedrooms	1	1	1
Bathrooms	1	1	2
Bedrooms imputed flag	1	1	0
Bathrooms imputed flag	0	0	0
Dist. to Cebu Business Park (m)	4,071	11,647	1,204
Dist. to Mandaue CBD (m)	1,652	11,014	2,220
Dist. to Mactan CBD (m)	5,778	6,618	5,300
Dist. to SRP (m)	11,037	16,316	8,107
Dist. to Talisay Tabunok (m)	15,335	19,077	11,868
Dist. to Consolacion (m)	8,436	10,714	8,317
Dist. to Naga City (m)	24,852	28,697	20,964
Dist. to Airport (m)	6,239	3,600	8,400
MCRAI composite	15.46	5.29	76.35
BIR zonal value (PHP)	21,150	5,000	30,000
Neighborhood price / spatial lag (PHP)	2,761,820	11,009,334	17,572,110
City: Consolacion	0	0	0
City: Lapu-Lapu City	0	1	0
City: Mandaue City	1	0	0
City: Minglanilla	0	0	0
City: Talisay City	0	0	0

Table D2

Feature snapshot — Houses stratum (24 model features).

Feature	Property 1	Property 2	Property 3
Floor / lot area (sqm)	50	184	170
Bedrooms	2	3	6
Bathrooms	2	2	3
Bedrooms imputed flag	0	0	0
Bathrooms imputed flag	0	0	1
Dist. to Cebu Business Park (m)	17,720	9,507	15,734
Dist. to Mandaue CBD (m)	21,656	7,194	19,670
Dist. to Mactan CBD (m)	25,260	12,497	23,274
Dist. to SRP (m)	10,612	17,241	8,626
Dist. to Talisay Tabunok (m)	7,031	21,540	5,996
Dist. to Consolacion (m)	28,072	8,650	26,086
Dist. to Naga City (m)	5,578	31,056	6,350
Dist. to Airport (m)	22,638	11,716	21,258
MCRAI composite	4.33	3.13	32.24
BIR zonal value (PHP)	7,000	10,250	7,000
Neighborhood price / spatial lag (PHP)	6,627,512	12,096,717	10,499,922
City: Consolacion	0	0	0
City: Lapu-Lapu City	0	0	0
City: Mandaue City	0	0	0
City: Minglanilla	1	0	1
City: Talisay City	0	0	0
Property type house and lot	1	1	1
Property type single detached	0	0	0
Property type townhouse	0	0	0

Table D3

Feature snapshot — Vacant Lot stratum (22 model features).

Feature	Property 1	Property 2	Property 3
Floor / lot area (sqm)	124	464	202
Dist. to Cebu Business Park (m)	13,136	11,810	15,540
Dist. to Mandaue CBD (m)	10,770	15,746	13,173
Dist. to Mactan CBD (m)	5,628	19,351	8,031
Dist. to SRP (m)	17,887	4,703	22,116
Dist. to Talisay Tabunok (m)	21,090	3,396	26,414
Dist. to Consolacion (m)	14,411	22,163	15,354
Dist. to Naga City (m)	31,763	11,637	35,931
Dist. to Airport (m)	3,913	17,093	2,520
MCRAI: Education	8.02	7.05	16.23
MCRAI: Grocery	2.49	4.61	10.04
MCRAI: Health	4.02	9.16	1.98
MCRAI: Hospitals	0.29	0.32	0.05
MCRAI: Recreation	6.83	2.19	9.70
MCRAI: Tourism	4.83	5.07	7.12
BIR zonal value (PHP)	10,000	9,000	5,000
Neighborhood price / spatial lag (PHP)	4,353,438	6,958,750	45,916,667
City: Consolacion	0	0	0
City: Lapu-Lapu City	1	0	1
City: Mandaue City	0	0	0
City: Minglanilla	0	0	0
City: Talisay City	0	1	0

Appendix E

OLS Diagnostic Plots

These plots support the OLS baseline diagnostics discussed in Chapters 3 and 6. They apply to the interpretable OLS comparator, not the deployed Random Forest, which does not assume constant error variance or freedom from collinearity.

Heteroscedasticity (Residuals vs. Fitted)

A funnel shape, where the spread of residuals widens or narrows across fitted values, indicates heteroscedasticity. This is why the OLS comparator was reported with HC3 robust standard errors rather than treated as a blocker for the deployed tree models.

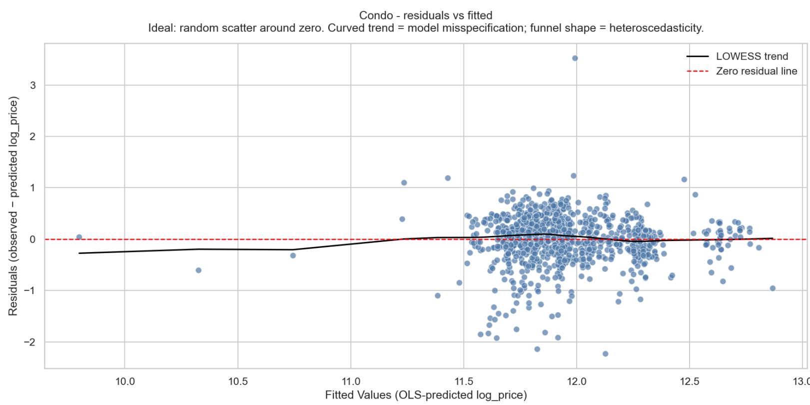


Figure E1

OLS residuals versus fitted values, condominium stratum.

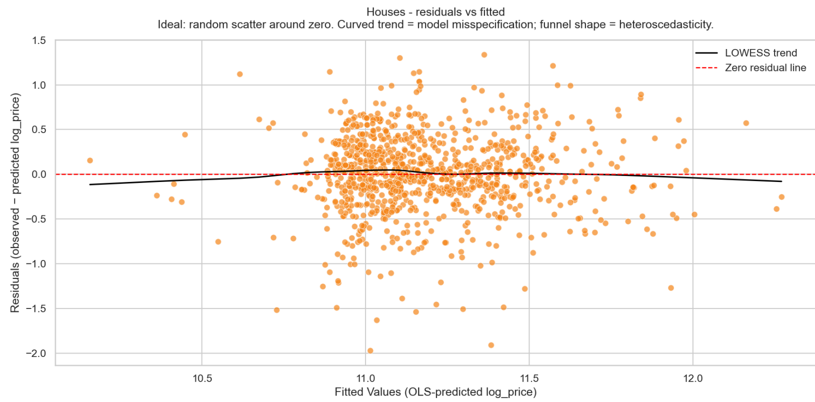


Figure E2

OLS residuals versus fitted values, house stratum.

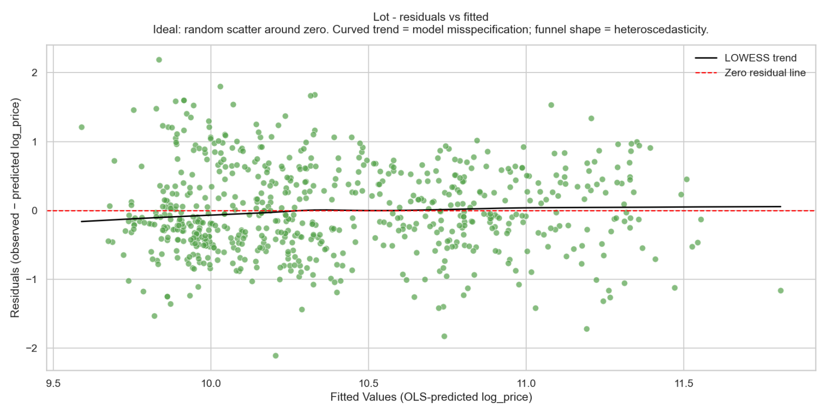


Figure E3

OLS residuals versus fitted values, vacant-lot stratum.

Multicollinearity (Variance Inflation Factors)

Variance inflation factors above the conventional thresholds (5 to 10) flag predictors whose information is largely carried by other features. These checks informed the per-stratum feature trimming described in Chapters 3 and 6.

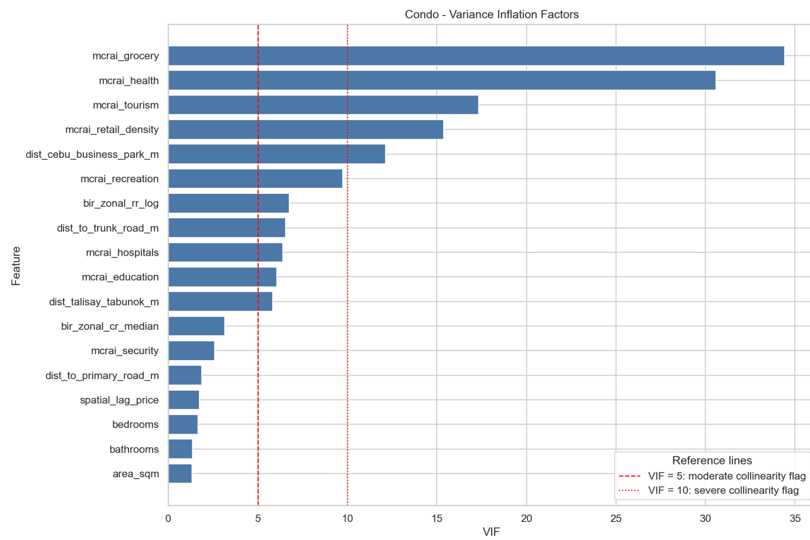


Figure E4

Variance inflation factors, condominium stratum.

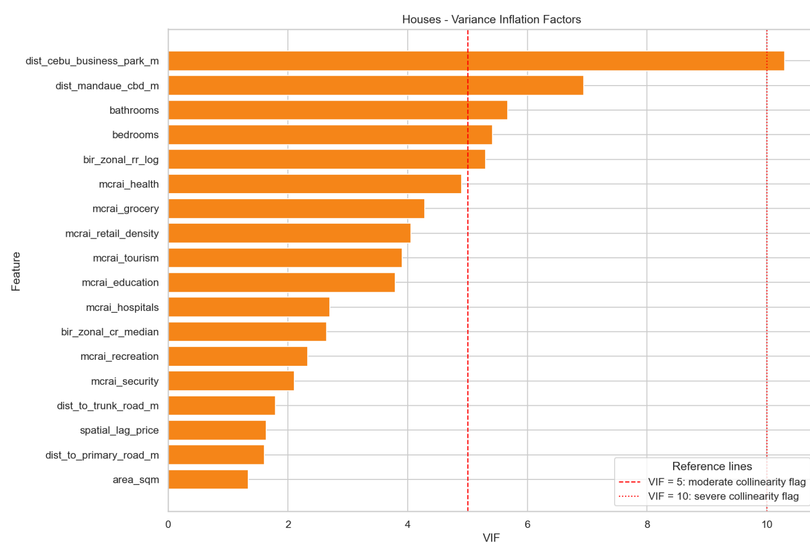


Figure E5

Variance inflation factors, house stratum.

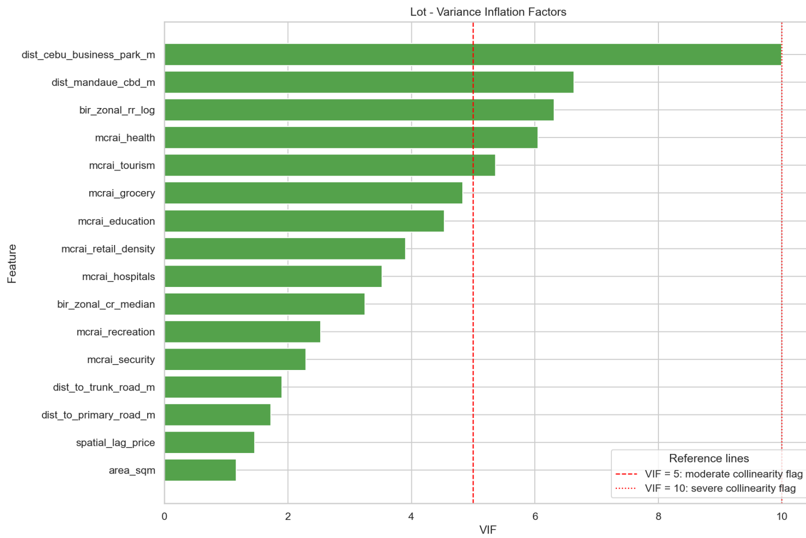


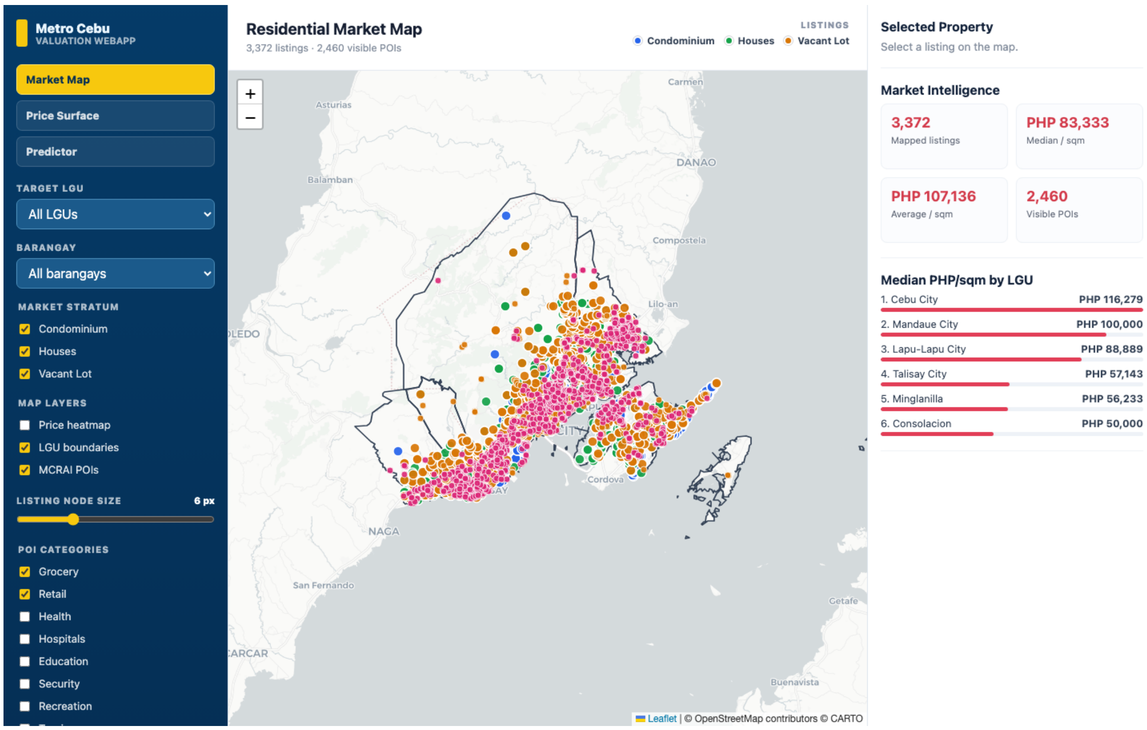
Figure E6

Variance inflation factors, vacant-lot stratum.

Appendix F

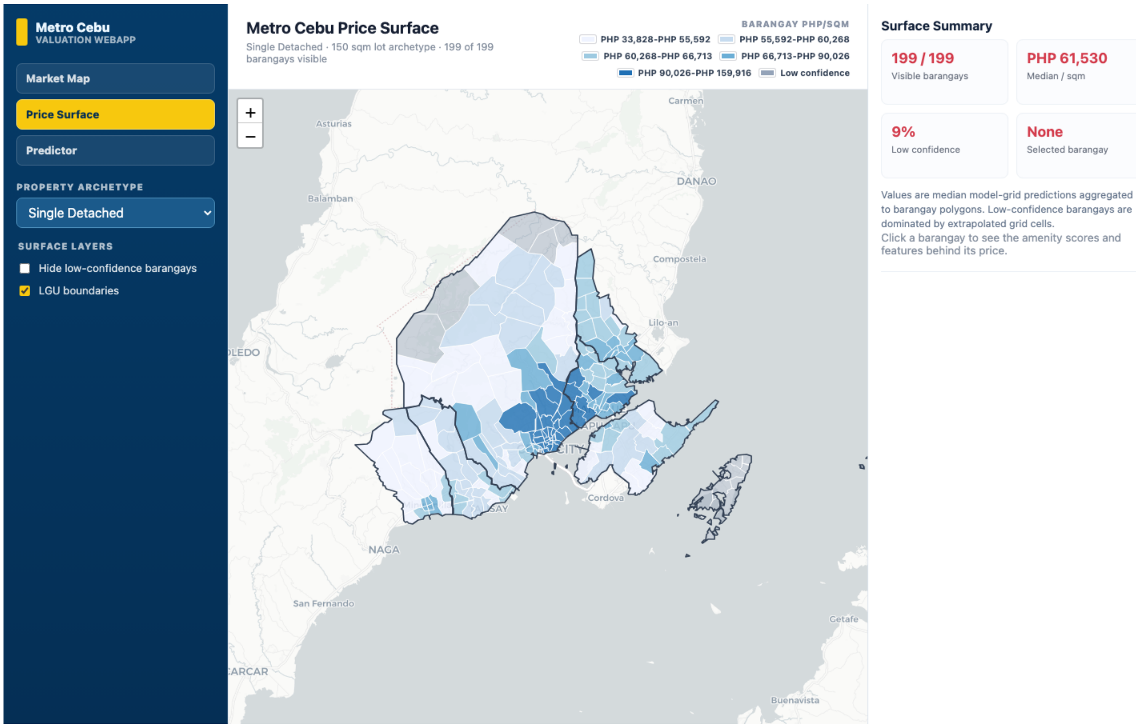
Web Application Screenshots

The prototype web-based decision-support application has three views, captured below. The backend runs the exact deployed Random Forest models, so the Property Predictor view shows a live prediction with its SHAP contribution breakdown.



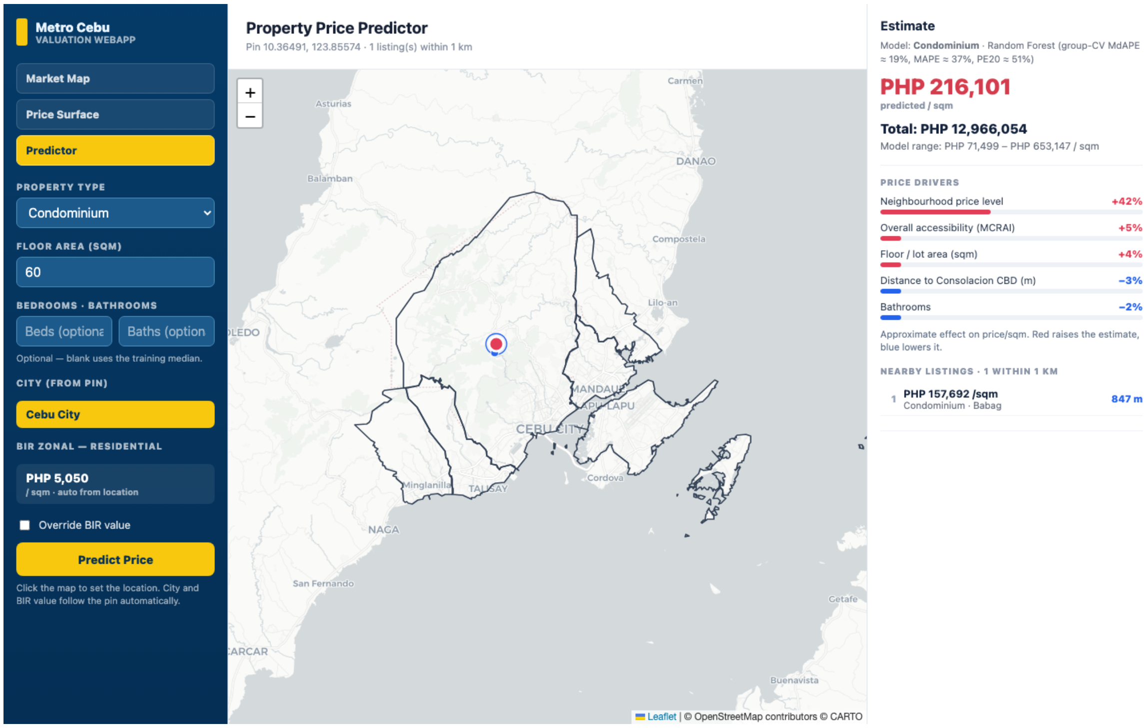
Picture 1

Prototype web application, Market Map view: open-market listings with filters and market-intelligence summaries.



Picture 2

Prototype web application, Price Surface view: the predicted price-per-square-meter surface aggregated by barangay.



Picture 3

Prototype web application, Property Predictor view: a property-level prediction with its SHAP contribution breakdown.